FishBot v0.5: A Self-Inflicted Deadlock, Its Measurement, and the Repair of the Ask Policy That Caused It

Live-ask gating, capacity-feasible declaration, and the retraction of an information-freeze claim in six-player Canadian Fish

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Repository: `https://github.com/dylann29/fish-optimization`
The diagnosis of §4 and the corrections of §8 were produced at commit `fe21e19`; the FishBot v0.5 policy of §5 is built on top of that commit. Diagnosis artifacts are in `research/v05/results/`.

Abstract

Canadian Fish, also called Literature, is a six-player imperfect-information card game in which two teams of three score only by declaring an exact allocation of a six-card half-suit among their own members, and in which nothing in the rules compels anyone to declare. FishBot v0.4 was evaluated against a panel of weaker policies and won decisively. Played against a copy of itself it does something else: over 300 mirror games, 39.04% of its asks name a card it can *prove* the target does not hold, 40.03% are exact repeats of an ask the same seat has already made, 34.33% of games contain a run of at least six consecutive provably dead asks — the longest run is 286 — and 10.44% of its declarations are wrong, rising to 58.59% for the declarations its own termination rule forces. Against FishBot v0.3 on the same instrument the dead-ask rate is 2.82%. The failure is invisible to the evaluation that shipped it because it is a strong-opponent phenomenon. This report diagnoses that failure, corrects the account of it given in the v0.4 study, and specifies the policy that repairs it. The deadlock is not an information freeze: it is a deterministic two-question cycle, present in 12 of 14 long games at the forensic seed, and across pooled seed banks no half-suit is locked to any team at any point in the cycle in 66% of long games. Where a half-suit *is* locked, further information is still obtainable — 68.5% of legal asks inside such a half-suit strictly raise a teammate's exact probability of the correct allocation, by a mean of 0.096 and up to 0.664 — so the claim in the v0.4 termination study that ownership monotonicity freezes allocation information is false, and is retracted here with the measurement that refutes it. The mechanism is arithmetic in the ask score: four ownership features that are not gated by hit probability pay up to +8.76 at hit probability zero, against $11.506 \cdot p$ for the hit-probability term, so a card the team already owns outscores a genuine chance at one it does not, and 99.3% of dead asks are chosen when a live ask was available. FishBot v0.5 restricts the candidate set to asks with strictly positive hard-consistent probability, replaces an unconstrained per-card allocation guess — responsible for *all* 281 forced-endgame declarations over 6,000 distinct mirror games being wrong — with a capacity-feasible one, and deletes an event-count rule that cashes a best candidate without inspecting its probability at a measured cost of 8.13 points against a mirror opponent. Refitted on a panel that for the first time contains an opponent of its own strength, and frozen, it takes the mirror dead-ask rate to 0.01%, exact repeats to 2.61%, games containing a dead run of six or more to 0%, declarations at or after the old horizon to 0, and declaration error to 2.07%, with the forcing rule switched off and no game reaching the action cap. None of that shows up as an aggregate win rate. Head to head it scores 51.61% [49.28–53.94] on the style bank and 50.77% over 5 further banks of 9,000 games; over a nine-style panel the two means are level, 83.65% against 83.57%; and on minimax regret over that panel FishBot v0.4 is marginally the better of the two, 2.06 against 1.67. Against three deceptive archetypes held out of the fit, the differences are larger and run both ways: +7.2 points on the withholding manoeuvre that prompted this work, +2.3 on a silent opponent, and -2.2 against a feint that manufactures a false ask-legality certificate. Each replicates in sign at 2 independent seed banks, and the withholding result replicates in size as well (+7.38 and +6.96); the silent one does not (+0.46 and +4.17). Seven further mechanisms are specified and not yet built. This report claims no equilibrium and no bound on exploitability, and, unlike the v0.4 study, it ran no exploitability probe at all.

1 Introduction

Canadian Fish, also called Literature, is a six-player card game of imperfect information: two teams of three, the whole deck dealt out, no communication outside the public record of play. On turn a player names a card and an opponent; if the opponent holds it the card transfers and the asker continues, otherwise the turn passes to the opponent asked. Cards taken this way score nothing. A team scores only by *declaring* — naming one half-suit and stating which of its own members holds each of its six cards — a correct declaration awarding the half-suit to the declaring team and an incorrect one to the opponents. §2 states every consequential rule choice in the dialect studied. Two of them govern everything in this report: a declaration may be made at any moment, and nothing in the rules ever obliges a player to make one.

FishBot v0.4 [2] was built on that structure. It computes an observer-conditioned posterior over the initial deal, scores asks with a twenty-feature linear function refined by a one-ply expectimax over a sixteen-feature value model, and decides declarations by comparing fitted value estimates. On its published evaluation bank it defeated FishBot v0.3 [1] in 75.07% of games. This report begins from the observation that the published evaluation could not have detected what FishBot v0.4 does against a strong opponent, and that the project owner met the resulting behaviour directly at the table: bots repeating one question at one opponent indefinitely, and then misdeclaring in bulk when the game was hurried to a close.

The measurement. §4 reports it. In mirror self-play, 39.04% of FishBot v0.4's asks are *provably dead*: the actor could prove from its own public deduction state that the target does not hold the named card. 40.03% of asks are exact repeats of an (actor, card, target) triple the same game has already seen. 34.33% of games contain a run of at least six consecutive provably dead asks; the longest such run is 286 asks. Against FishBot v0.3 on the identical instrument the dead-ask rate is 2.82% and the longest run is 5. The pathology is a strong-opponent phenomenon, which is precisely why a panel of weaker policies did not expose it, and why a competent human — who is closer to the mirror case than FishBot v0.3 is — meets it and FishBot v0.3 does not.

Five claims this report makes, and one it retracts.

1. **The deadlock is an ask-policy fixed point, not an information freeze.** The v0.4 termination study explains non-termination by appeal to the locked-half-suit theorem, asserting that no further information about a locked half-suit's allocation can ever arrive. Both halves of that account fail. The observed deadlock is a deterministic two-question cycle: in 12 of 14 long games the entire dead run consists of exactly two distinct (actor, card, target) triples, and in 66% of long games no half-suit is locked to any team at any point during the run. Independently of that, the informational claim is false on its own terms: 68.5% of legal asks inside a half-suit a team does in fact own strictly increase a teammate's exact probability of the correct allocation, mean +0.096, maximum +0.664, and 59.9% strictly lower its exact marginal entropy. §8 states the retraction.

2. **The cause is arithmetic, and it is in the ask score.** Four ownership features are not gated by hit probability. Evaluated at the compiled weights they pay up to +8.76 at $p = 0$, against $11.506 \cdot p$ for the hit-probability term, and the only explicit information term in the score, the binary entropy of the ask outcome, carries weight -2.6534 — a negative value-of-information term, so a certain miss is rewarded for being certain. 99.3% of the dead asks are voluntary: a live ask existed and was passed over.
3. **Terminating by force is a strong-opponent tax, and terminating by patience does not terminate.** A configuration whose event-count escalation never fires beats the shipped one by 56.2% (deal-clustered 54.93–57.47%) and declares at 98.40% accuracy against 85.66%, but reaches the action cap in 22.5% of mirror games. The escalation's second stage, which cashes a best candidate without inspecting its probability at all, costs 8.13 points against a mirror opponent and exactly nothing against every weaker style tested. Both settings of the dial are bad; the missing capability is an ask rule that terminates the cycle by choosing informative asks rather than an event counter that ends the game by guessing.
4. **The repairs that follow are keyed to measurements, and the ones that failed are recorded as failures.** §5 specifies ten mechanisms; three are built and seven are designed. It also records four plausible fixes that were measured and rejected — a time-varying holding cost, deleting the policy prior, a turn-transfer willingness ladder, and confidence-ranked declaration arbitration — so that they are not rebuilt.
5. **The repair removes the failure mode, and does not show up as a win rate.** The three built mechanisms were refitted, on a panel that for the first time in this project's history contains an opponent of the agent's own strength (§6), and frozen. In the mirror, the dead-ask rate goes from 39.04% to 0.01%, exact repeats from 40.03% to 2.61%, games containing a dead run of six or more from 34.33% to 0%, and declaration error from 10.44% to 2.07% — with the event-count escalation switched off. Head to head FishBot v0.5 scores 51.61% [49.28–53.94] on the style bank and 50.77% over 5 further banks; across the nine-style panel of §7 the means are 83.65% against 83.57%; and on minimax regret over that panel FishBot v0.4 is marginally the better of the two, 2.06 against 1.67. Where the two policies do separate is against deception: +7.2 points on the withholding manoeuvre that prompted this work and +2.3 on a silent opponent, against -2.2 on a feint that manufactures a false ask-legality certificate. Each replicates in sign at 2 independent seed banks; the withholding and feint results replicate in size too, the silent one does not (+0.46 against +4.17).

What this report does not claim. No result here is an equilibrium result, and nothing here bounds exploitability from above. The v0.4 study ran an exploitability probe; this one ran none at all, so FishBot v0.5 makes no robustness claim that a best response could contradict, and the comparison of the two policies is not a comparison of their exploitability. Following the project's standing evaluation preference, no aggregate figure above appears without the per-opponent breakdown and an explicit worst case beside it: over the twelve styles tested, and on the per-style banks those figures come from, FishBot v0.5's worst case is 50.96% against the feint and FishBot v0.4's is 50.00% against a copy of itself, which is the worst case for any policy in this family. The fit behind those figures is a single round from a single base seed, against FishBot v0.4's 5 rounds, and the sixteen-coefficient value model inside the policy was not refitted at all. Seven of the ten mechanisms of §5 are specified and unbuilt, so nothing here measures the design this report proposes — only the three parts of it that exist.

2 The game and the rules dialect studied

Canadian Fish (also called Literature) is played by six players in two teams of three, seated so that the teams alternate around the table. The deck is 54 cards, partitioned into nine *half-suits* of six cards each: the low band $\{2, 3, 4, 5, 6, 7\}$ and the high band $\{9, 10, \mathrm{J}, \mathrm{Q}, \mathrm{K}, \mathrm{A}\}$ of each of the four suits, together with a ninth half-suit containing the four eights and the two jokers. The dealer is chosen at random, each player is dealt nine cards, and the player to the dealer's left leads. A half-suit leaves play when it is claimed; the team that takes at least five of the nine wins the game.

The dialect below is the one supplied by the project owner as the rules of record, and the engine's `Rules` defaults match it on every point checked. Published descriptions document mutually incompatible variants of several of these choices [3]; what follows is one explicit selection, not a uniquely canonical rule set. It matches the strategy corpus this work draws on [4], and the contested choices are exposed as engine switches so that their effect can be measured rather than assumed.

Asking. On their turn a player asks one opponent for one named card. The ask is legal only if the named card lies in a half-suit that is still live, the asker holds at least one *other* card of that half-suit, the asker does not hold the named card, the target sits on the opposing team, and the target still holds at least one card. If the target holds the named card they must surrender it and the asker retains the turn; otherwise the turn passes to the target. Every ask, its answer, every resulting transfer, and every player's card count are public. The rules additionally license two public queries that the engine models implicitly because both quantities are already in the public state: the content of the previous two asks, and any player's remaining card count.

Declaring. A claim on a half-suit is a *declaration*: an exact allocation naming, for each of the six cards, which member of the declaring player's own team holds it. If every one of the six assignments is correct the declaring team is awarded the half-suit; *any* inaccuracy at all awards it to the opposing team. Either way the six cards leave play, the half-suit becomes inactive, and the turn returns to the player who held it before the declaration. A declaration is legal at any moment, including during an opponent's turn. The declarer need hold no card of the half-suit — indeed need hold no cards at all — and may not declare on behalf of the opposing team.

Running out of cards. A player with no cards leaves the asking cycle: they can neither ask nor be asked, though they may still be named as the holder of a card in a teammate's declaration and may still declare. If the turn reaches a cardless player — possible only after a declaration — the turn passes to a teammate who still holds cards. When an entire team is cardless, the other team must declare every remaining half-suit with no further asking, and may misdeclare and thereby hand half-suits to the cardless team.

2.1 Two places where the engine is more restrictive than the rules

Both are coordination channels the rules permit and the engine does not use. They are stated here because §5 reports that both were measured and neither is worth building, which is a result about the game and not only about the implementation.

1. **Turn transfer.** The rules say that when a cardless player must pass the turn and both teammates hold cards, the team may openly strategise about which teammate receives it, but may share nothing beyond their willingness to receive it. The engine instead has the cardless player choose unilaterally from its own belief, and solicits no willingness bit. A genuine multi-candidate decision arises 0.148 times per game; §5.4 reports what the channel is worth.

2. **Declaration arbitration.** The rules do not say who acts first when two seats offer a declaration in the same instant. The engine resolves it by lowest seat, deliberately, to avoid comparing private confidences across seats. §5.4 reports the measured cost of that choice.

2.2 Engine settings, and the switches that change them

- **Deck:** 54 cards, nine half-suits, nine per player (`--sets=8` gives the 48-card, eight-half-suit form).
- **Declaration timing:** any seat may declare at any moment, in or out of turn, holding cards or not (`--no-out-of-turn`, `--no-cardless-declare`).
- **Misdeclaration:** the half-suit is awarded to the opposing team; published variants differ here [3].
- **Simultaneous declarations:** the lowest seat index that offers a declaration acts first — a modelling choice, not a rule (`--arb=low|high|turn`).
- **Forced endgame:** an eight-level willingness ladder selects the declarer when one team is cardless (`--legacy` gives the two-level v0.3 ladder). §4.7 reports that in the shipped policy this ladder is inert.
- **Action cap:** 400 asks, any residue adjudicated neutrally by majority physical holding (`--maxasks`).

No prearranged signalling conventions are available to any policy in the configuration measured here, and team communication is limited to the willingness bit above and the successor choice made by a cardless turn-holder. The literature records that pre-agreed conventions are explicitly forbidden in this game [4] while also documenting a named convention in real use [3], and a pair of policies trained together has an implicit pre-agreement by construction; §5.3 states how FishBot v0.5 handles that tension.

The action cap is a backstop, not a rule. Two sufficiently patient policies can stall indefinitely, and this report is largely about a configuration that does. Its incidence is therefore reported with every experiment rather than assumed to be zero: it is 22.5% for the unforced mirror configuration of §4.6 and zero for the shipped FishBot v0.4 configuration, which avoids it by forcing declarations that §4.6 shows are 58.59% wrong.

3 Related work

The v0.4 study surveyed inference, search, team equilibria and signalling, and that survey stands. This section covers what the present report additionally depends on, and it is organised around a gap: a coverage sweep of the eight reviews assembled for v0.4 finds no occurrence of *war of attrition*, *minimax regret*, *distributionally robust*, *Dynkin*, *restricted Nash*, *data-biased* or *safe opponent exploitation*, and one occurrence of *deadlock*. Those are exactly the two bodies of work this report needs: the theory of voluntary stopping when both sides may wait, and the theory of performing well against the worst opponent style rather than the average one.

3.1 Fish and Literature

The published material on this game remains human and informal. McLeod's Pagat entry [3] is the canonical rules source and records the points on which dialects differ; Develin's chapter [4] is the most detailed strategy writing located, and supplies two observations this design uses — that an ask teaches the opponents as much as it teaches a teammate, and that pre-agreed conventions are outside the game. A repeat search on 22 August 2026 — the arXiv API for `"Canadian Fish"`, `"Literature card game"` and `"half-suit"`, returning zero results, plus a web sweep that surfaced only adjacent games — reproduced the v0.4 finding. Every novelty statement in this report therefore takes the form "we found no prior published work that ... " and should be read against that scope. The one agent located, Somani's `literature` [5], plays the four-player 48-card variant and publishes no strength numbers; a closed-source mobile application advertises six-player bots with no stated methodology [6]; Literature is not among the environments packaged with RLCard [7].

3.2 Voluntary stopping when waiting is free

The declaration decision in Fish is a multi-player, nonzero-sum, undiscounted optimal-stopping game: each side chooses when to claim, the payoff depends on who claims first and whether the claim was right, and waiting costs nothing in the rules. Games of this shape are known to be badly behaved. Christensen, Lindensjö and Neumann [8] record that for undiscounted Dynkin games only ϵ-equilibria are established in general, with counterexamples showing that Nash equilibria need not exist even for deterministic stationary rewards, and that this survives enlargement to randomised stopping times. Hamadène, Hassani and Morlais [9] prove existence of an ϵ-Nash equilibrium for the N-player discrete-time case by a constructive algorithm, under a payoff structure that is the Fish structure: payoffs depend on the set of players who stop at the first stage at which anyone stops. These are results for general Markovian reward processes rather than theorems about Fish, so the correspondence is one of structure and is used here to name the missing ingredient, not to assert a property of the game.

The classical model of mutual refusal to move is the war of attrition. Hendricks, Weiss and Wilson [10] characterise the complete-information continuous-time case and find a continuum of equilibria including arbitrarily long delay: delay is an equilibrium phenomenon that a symmetric deterministic pair will select, and FishBot v0.4's mirror match is exactly symmetric and exactly deterministic. The constructive converse is Mguni [11]: impose a strictly positive cost on each action and the stochastic game acquires a unique value and a Markovian saddle point at strategically chosen stopping times. That is the cleanest published statement of the natural fix, and §4 reports that in this game it does not work — the cost is a mean-shifter that leaves the tail of the distribution untouched.

There is a reason to be careful with holding costs beyond their measured ineffectiveness. Admati and Perry [12] show that delay is used as a signalling device precisely because it is a more efficient signal of strength than the alternatives, and that the delay does not vanish as the interval between moves goes to zero. Applied to Fish this is Farrell–Gibbons two-audience cheap talk [14] on the *timing* channel: if how long a seat waits before declaring depends on its private confidence, three opponents can read that confidence. Any holding cost must therefore be a deterministic function of public state, or be deliberately randomised.

We found no substantive literature on self-inflicted non-termination in game-playing agents. Three distinct query formulations returned only anecdote, resolved in every case by an arbitrary move cap — which is what FishBot v0.4 does, and what §4 measures the cost of. That absence is itself a result: voluntary-claiming games are an under-studied class and Fish is a clean instance of one.

3.3 Information-seeking actions, and what an ask leaks

An ask in Fish is simultaneously a request for material, a certificate to one's teammates and a disclosure to one's opponents. The v0.4 review formalised the trade-off as a per-decision secrecy rate in the Wyner [13] sense — the mutual information the action carries to teammates minus the mutual information it carries to opponents — and built none of the four instruments it listed. FishBot v0.4 contains three information terms in its ask score and all three are negative; §4.5 gives the arithmetic.

Two results bear directly on the repair. Alvim et al. [15] model information flow as a game in which the defender's utility is a convex function of the *distribution* over its actions rather than the expected utility of the action played, and show that a Nash equilibrium nevertheless exists. The consequence for Fish is sharp: a deterministic argmax ask policy has maximal leakage by construction, so no reweighting of leak-penalty features can reduce leakage below the deterministic bound. Reducing it requires randomising over near-tied asks. The second consequence is a caveat — because utility is convex in the mixed strategy, the leakage game is not bilinear and standard regret matching does not apply unmodified.

Lervold, Peterson and King [16] appear to be the only published work that places an explicit information-gain incentive inside an information-set tree search, reporting improvement over baseline and over a cheating search in a hidden-information multi-action wargame. We were unable to obtain the full text, so the finding is used here and the formula is not. Serrino et al. [17] integrate deductive reasoning directly into vector-form counterfactual regret minimisation for five-player Avalon; it is the closest published system to Fish's structure, a hidden-role adversarial team game in which every action is public and its informational content is the whole game, and it is the existence proof that hard deduction can live inside a solver rather than beside it.

3.4 Online opponent modelling, and robustness to being modelled

FishBot v0.4 models the other five seats with two global scalars, identical for every opponent and never updated during a game (§4.8). The literature on doing better within a single short episode is mature and was absent from the v0.4 corpus. Ganzfried, Wang and Chiswick [18] give the only opponent-modelling result located that is explicitly in the more-than-two player regime. Rosman, Hawasly and Ramamoorthy [19] pose exactly this problem — select online, from a library of policies, a response to a novel task instance, under a short episode — and a Fish game is 144 public events long. Smith et al. [20] learn a value function against each pure opponent and combine them for any mixture without retraining, with a classifier that refines the mixture weights during the episode.

The robustness half matters more here, because the behaviour that prompted this work is an opponent deliberately feeding a model. Johanson, Zinkevich and Bowling [21] give the shape of the dial: solve a game in which the opponent plays a fixed model with probability p and is free otherwise, so that $p = 1$ is a best response and $p = 0$ an equilibrium, and intermediate p traces an exploitation/robustness frontier. Johanson and Bowling [22] set p per information set as a function of how much data is actually available there — the right refinement for Fish, where the per-half-suit evidence about a given opponent is a handful of observations, and the one that makes a deliberately silent opponent fall back to the population prior instead of being misread. Ganzfried and Sandholm [23] guarantee at least the game value per period while still exploiting mistakes; Müller et al. [24] give the modern bandit-feedback formalisation of the same idea. Yang et al. [25] handle opponents that are themselves running Bayesian opponent models, and Milec, Kovařík and Lisý [26] adapt to sub-rational opponents beyond a depth limit while remaining robust against rational play.

3.5 Worst case rather than mean case

The standing evaluation preference of this project — report the per-opponent breakdown and an explicit worst case, never an aggregate alone — has a matching equilibrium concept. Aggarwal and How [27] define a probabilistically robust minimax-regret equilibrium for adversarial team games under asymmetric information, minimising worst-case regret over a high-confidence subset of the type space rather than over the whole of it, and solve it inside a robust double-oracle loop. Translated to Fish, the type is the opponent style, the high-confidence subset is the style set actually built, and the prescription is to optimise worst-case regret rather than either the mean or the absolute worst case — promoting a reporting rule to a training objective. Two practical notes come with it. Xu et al. [28] observe that computing max-regret requires the per-style optimum first, which in this setting is one arena run per style, so minimax regret is available as a post-hoc selection criterion over checkpoints and not only as a training objective. Beukman et al. [29] document the failure mode of a naive worst-case objective: once the bound is attained everywhere, learning stagnates because the objective stops distinguishing policies — which is a direct warning against a raw minimum over a small style set. Vinitsky et al. [30] report that a single adversary produces exploitable policies and replace it with a sampled population.

3.6 Team games, and the concepts this report does not compute

Three teammates share a payoff, hold different information and cannot communicate, so the target concept is a team-maxmin equilibrium with correlation [31]. Nothing here computes one. Two recent results scope the ambition. Anagnostides et al. [34] show that computing Nash equilibria in team zero-sum games is PPAD-complete, and remains so at inverse-polynomial precision and with two-player teams; that concerns the uncorrelated equilibrium, which is what three independently acting seats without a shared per-episode seed actually implement. Liu et al. [33] identify a concrete defect in Team-PSRO [32]: its policy sharing cannot cover the joint policy space in *heterogeneous* team games where teammates play distinct roles, yielding higher exploitability. Fish seats are heterogeneous in the relevant sense, since turn flow and distance to the next opponent differ by seat, so any future population loop for this game should be the heterogeneous variant.

Two design questions in §5 are the Hanabi-style zero-shot coordination questions [35] in another game: how much a policy may rely on its partner running the same blueprint, and how to keep a convention from degenerating into a private code. Off-belief learning [37] and other-play [36] are the reference treatments, and the level of grounding is the natural explicit knob for the partner-regime distinction of §5.2.

3.7 Determinization, and why it is not the foundation used here

Perfect Information Monte Carlo — sample hidden cards consistently with the public history, solve the resulting perfect-information game, average [38] — is unsound however many worlds are drawn [39], and is degenerate in Fish for a structural reason as well: a clairvoyant player never fails an ask, so almost every reachable position carries the same perfect-information value and the average collapses to choosing the ask most likely to hit. The v0.4 study gives the collapse in full. It is worth restating here because the collapse target — “choose the ask most likely to hit” — is a policy that would not have deadlocked. FishBot v0.4’s pathology is not that it is a hit-maximiser; it is that its ownership features outbid the hit-probability term at $p = 0$ (§4.5). The policy-aware reweighting of a rules posterior that the Skat literature uses [41, 42, 40] is the correct frame for the prior that §4.8 examines, and the counting machinery underneath [43, 44] is unchanged from v0.4.

4 Diagnosis: what FishBot v0.4 does against a strong opponent

This is the central section of the report. It establishes four things in order: what the failure is, that it is not the failure the v0.4 study described, what causes it, and what each contributing defect costs. Every number is produced by an instrument described in §4.1 and carries the artifact it came from.

4.1 The instrument

The measurements below come from a diagnostic command added for this study, `fish pathology`, together with five probes that re-derive its headline quantities independently. Three definitions do the work.

Definition 1 (Provably dead ask)**.** An ask of card c from target t by actor a is *provably dead* if a’s own public deduction state establishes that t does not hold c: either c’s holder is already known and is not t, or t has been excluded from c’s candidate set by a certificate. The predicate is a deduction from

Table 1: FishBot v0.4 measured on the same instrument against itself and against FishBot v0.3. Mirror: 300 games, seed 31. Held-out: 600 games, seed 90210. Source: `research/v05/results/P0-v04-pathology.md`.

	mirror	**vs. FishBot v0.3**
events per game	144	101.7
ask hit rate	34.25%	54.9%
provably dead asks	39.04%	2.82%
exact repeat (actor, card, target)	40.03%	5.29%
longest dead run	286	5
games with a dead run ≥ 6	34.33%	0%
starved turns	0.2%	0.03%
asks inside a half-suit the actor's team already owns	16.46%	11.7%
declarations wrong	10.44%	8.96%
declarations at or after the forcing horizon	384	0
of those, wrong	58.59%	—
forced-endgame declarations wrong	100%	100%

the actor's own hand and the public record. No posterior and no policy prior enter it, so a policy that filters on it cannot acquire a modelling error — it can only lose moves whose probability of success is exactly zero.

Definition 2 (Dead run)**.** A maximal sequence of consecutive asks in one game, every one of which is provably dead in the sense of Definition 1.

Definition 3 (Starved turn)**.** A turn at which *every* legal ask is provably dead, so the actor has no live alternative. Starved turns separate a policy that chooses dead asks from a policy that is forced into them.

Two populations are used throughout: *mirror* play, in which both teams run the same policy, and the *held-out* contrast against FishBot v0.3, which is the population on which FishBot v0.4 was evaluated for publication.

4.2 The baseline, and why the published evaluation could not see it

Table 1 is the whole problem in one place. On the population FishBot v0.4 was published against, it is a clean policy: 2.82% dead asks, no game containing a dead run of six, no game reaching the forcing horizon at all. Against a copy of itself, two in five of its asks name a card it can prove the target does not hold, two in five are exact repetitions, and a third of games contain an extended run of guaranteed misses. The two columns differ by a factor of fourteen in dead-ask rate. A policy panel of weaker opponents cannot detect this, and a competent human opponent is closer to the mirror column than FishBot v0.3 is — which is how the failure was first reported, from live play, and not from any experiment in the v0.4 study.

The pathology is not caused by the approximate inference path. Substituting the exact reference belief reduces dead asks from 39.04% to 28% and leaves the longest dead run at 280. The belief reports the hit probability correctly and the score prefers the move anyway: this is a policy defect, not an inference defect.

4.3 The deadlock is a two-question cycle

Of 120 mirror games scanned in the forensic probe, 14 exceed 300 events. In those games the mean longest dead run is 245.1 asks, and the mean number of distinct (actor, card, target) triples inside that run is 2.14: in 12 of 14 the entire run consists of exactly two questions, alternating, each asked well over a hundred times. Two seats on opposite teams ask each other the same guaranteed-miss question until an event counter stops them.

The reason a deterministic policy can do this is that the position is a fixed point. Sampling each dumped run at its onset and at +40 and +80 events, every observer's exact mean per-card certainty is identical to the last digit at all three samples, as are the hand counts, the unresolved counts and the score. The only quantity that changes is a counter: each repeated ask appends a fresh ask-legality certificate that duplicates one already held, so the stored certificate list grows without bound while the posterior it induces does not move. Both policies re-derive the same argmax from the same information state, and the cycle closes.

4.4 The deadlock is not an information freeze

The v0.4 termination study attributes non-termination to the locked-half-suit theorem, stating that such a half-suit is frozen and that, for the same reason, no further information about its allocation can ever arrive. That account fails in two independent ways, and §8 retracts it.

First: the deadlock is mostly not in a locked half-suit. A ground-truth lock census taken at *every* event of the dead run — not only at its ends — finds no half-suit locked to any team at any point in the run in 41 of 62 long games pooled over four seeds (66%). Measuring the stronger and more relevant condition, only 13.0% of dead-run asks sit inside a half-suit that is locked at that moment, and only 3.2% of long games have their entire cycle inside a lock. In the majority of deadlocks the theorem has nothing to bite on: at the onset of one traced run all nine half-suits are live and genuinely split, 51 of 54 legal asks are not provably dead, and the policy ranks a guaranteed miss first and then repeats it 142 times.

Second: where a half-suit *is* locked, information still arrives. This is a statement about the rules before it is a measurement. An ask inside a half-suit locked to the asker's own team is legal — the set is active, the target is a live opponent, the asker holds another card of the set and lacks the one named — and the deduction machinery publishes the asker's own exclusion regardless of whether the ask succeeds. "The asker does not hold this card" is public, permanent information about the allocation of a locked half-suit, and with six cards split among three teammates it frequently resolves a card outright.

Pooled over three independent seeds and 2,880 legal asks inside a half-suit the asker's team in fact owns: 68.5% strictly increase a teammate's exact probability of the correct allocation, by a mean of +0.096 and a maximum of +0.664. Re-measured with two statistics that avoid the disputed estimator entirely — the combinatorial support size of the half-suit's cards, and the Shannon entropy of the exact marginals — 59.9% strictly lower the entropy, by a mean of 0.325 nats. Asks *outside* the locked half-suit shrink its support in 0 of 20,898 cases at any seed, which is the expected structural result and confirms that the certificate lands where it is aimed. Playing such asks greedily, a team's best allocation probability reaches certainty in 82.9% of attempts in a median of 4 asks.

Ownership monotonicity therefore does not imply informational monotonicity, and the two must not be merged. The v0.4 *paper* states this correctly as an observation; the v0.4 termination artifact does not, and it is that artifact's claim which §8 withdraws.

The leak, priced. These certificates are public, so opponents read them too. Over the same ladders, the team's allocation probability on the half-suit it is trying to cash rises by +0.780 while the opponents' best locked-half-suit allocation probability rises by +0.021 — an exchange rate of about

37 : 1 in the sender's favour. The opponents' gain is indirect, arriving through capacity coupling rather than through the locked half-suit itself, which they can never take.

What the channel costs, and the qualification that matters. A guaranteed miss donates the turn. In FishBot v0.4's own value function a donated turn is worth 0.128 half-suits of final differential, and empirically the receiving team's take from a miss inside a dead run is 0.002 half-suits, with 99.6% of such donations yielding the opponents nothing at all. The qualification is important and was overstated in the first pass of this analysis: the lock census that selects these asks is a ground-truth statistic. At the deadlock states examined, the exact probability an owning team assigns to its own ownership has mean 0.068 and maximum 0.342, and not one of the 36 (observer, half-suit) pairs clears the 0.999 bar at which ownership would be provable. A FishBot v0.5 ask rule therefore *cannot* condition on "a half-suit my team provably owns"; the same conclusion is reached independently in §4.10, where the provable version of the condition holds at 0.34% of mirror decisions. The channel is real and it must be priced against ownership probability, not gated on a certainty nobody has.

4.5 The cause: ownership features that are not gated by hit probability

The ask score is a twenty-feature linear function refined by a one-ply expectimax. Four of its features measure ownership progress and none of them is multiplied by the probability that the ask succeeds: own-set progress (+1.688), team control (+2.133), lock completion (+4.071) and the completion bonus (+1.428). Evaluated at the compiled weights they pay up to +8.76 at $p = 0$, against $11.506 \cdot p$ for the hit-probability term. A card the actor's whole team already owns therefore outscores a genuine chance at a card it does not.

Three further features push the same way. The score's only explicit information term is the binary entropy of the ask outcome, at weight −2.6534: the ask whose result would resolve a full bit is penalised most, and the ask whose result is already known is penalised not at all. This is a negative value-of-information term in the literal sense — the policy does not merely ignore information, it charges for it. A second feature rewards re-asking in the half-suit last asked in (+1.270), which is a direct mechanism for the 40.03% exact-repeat rate. A third scales the exposure penalty by $(1 - p)$ and is therefore maximised exactly when the ask is certain to miss.

Evaluating the compiled weight vector at two representative feature vectors, a provably dead ask inside a half-suit the team already owns outright scores 7.44 while a maximally informative fresh ask scores 5.09. That is an illustration rather than a game measurement — the shipped policy adds an expectimax term, so the argmax is not the linear argmax — but the decomposition of real decisions agrees with it. At the traced deadlock onsets the gap between the dead ask ranked first and the best live ask decomposes into the same three contributions every time: entropy, exposure-on-miss and target hand size outvote the hit-probability term.

Two facts establish that this is a choice and not a constraint. Starved turns, at which no live ask exists, are 0.2% of asks; 99.3% of provably dead asks are made with a live alternative available. And the most informative legal ask available at a deadlock state ranks, in FishBot v0.4's own score, 17.6 out of roughly forty on average. At the 42 deadlock states examined the ask actually played moved no teammate's exact posterior in every case, while an ask that would have moved it existed in every case. The ask rule is not slightly mispriced on information; it does not contain the term.

Causal isolation. Filtering provably dead asks out of the candidate set, with nothing else changed, collapses the longest dead run from 289 to 1, takes games containing a dead run of six from 32% to 0%, takes declarations at or after the horizon from 474 to zero, takes misdeclarations from 10.9% to 1.9%, and scores 49.75% against the shipped policy — neutral. The single change removes the pathology at no measured cost in strength, which identifies it as the load-bearing defect.

Table 2: The two settings FishBot v0.4 has, and what each costs. Sources: `research/v05/results/P0-v04-pathology.md`, `research/v05/results/P0b-forcing-dilemma.md`.

	forcing (shipped)	**patient**
declarations wrong	10.44%	2.1%
games ending by action-cap adjudication	0%	22.5%
provably dead asks	39.04%	51.2%
longest dead run	286	379

Removing the repetition without pricing information is a different matter, and it fails. Forbidding exact repeats alone terminates every game, taking events per game from 140.6 to 101.1 and the longest dead run from 284 to 4, but it loses to the shipped policy at 42.2% over 2,000 games. Breaking the cycle is not sufficient; the candidate set has to be restricted to asks that can actually do something.

4.6 Terminating by force, and terminating by patience

FishBot v0.4 terminates deadlocks with a graduated escalation keyed on the public event count. Raising the horizon so that the rule never fires is worth 56.2% against the shipped configuration (deal-clustered 54.93–57.47%, $n = 1{,}500$), with the patient side declaring at 98.40% accuracy against the forcing side's 85.66%. But pure patience does not terminate: the unforced mirror reaches the action cap in 22.5% of games, with 51.2% of asks provably dead and a longest run of 379. Table 2 states the dichotomy.

Note the third row. In the patient mirror more than half of all asks are already provably dead: the turns are being burned either way. They are simply not being spent on asks that emit useful certificates. That is the sense in which the project owner's judgement — that a risky ask is better than a guaranteed-zero declaration — is a statement about the ask policy and not about the declaration rule.

The second stage of the escalation is an unconditional cash. Past a second horizon the rule returns "declare" before inspecting the allocation probability at all, and simultaneously zeroes the team-ownership floor and the marginal pre-gate. The consequence is measurable in the policy's own stated confidence: bucketing declarations by the probability the policy itself attached to them, 130 declarations in 200 games are made at a self-assessed probability below 0.1 and every one of them is wrong. Pre-horizon declarations are wrong 1.83%; at or after the horizon, 66.54%. Disabling the second stage is worth 8.13 points against a mirror opponent and changes nothing against FishBot v0.3, the detective, the lockout policy, the hunter or the bluffer — differences of at most a third of a point, in both directions. It is a pure strong-opponent tax.

One hypothesis about this rule did *not* hold and is recorded so that it is not repeated: the horizon does not fire on healthy long games. Every one of the 46 games (23.0% of 200) that reached it was genuinely deadlocked, and raising the deadlock threshold to a run of 80 leaves the split unchanged. The cost is not when the rule fires but what it does when it fires.

4.7 The forced endgame: an allocation with posterior probability zero

When one team is cardless the other must declare every remaining half-suit. Over 6,000 distinct mirror games, all 281 forced declarations are wrong. An independently instrumented replay that shares no code with the diagnostic probe and recomputes each declaration's correctness from the opening deal reproduces the result at five further mirror seeds, disagreeing with the engine's own flag on none of the declarations it re-scored. This is not a hard decision problem and not an accounting artefact; it is one mechanical defect with a deterministic consequence. The routine that names the allocation picks each card's owner by an independent per-card argmax over ownership marginals, with no capacity constraint. In every observed forced endgame the truth splits the unresolved cards across two teammates and the argmax piles them onto one. The named allocation therefore violates the declarer's own capacity constraint, has exact posterior probability zero, and loses the half-suit with certainty. All 281 violate capacity; all 281 have exact posterior probability zero; the declarer's own stated confidence is exactly zero in all 281; and the best *feasible* allocation would have been correct in 40.6% of them. The policy misses by the minimum possible margin every time — one card in 92.2% of them, two in the remainder, never more.

The same defect disables the willingness ladder that selects the declarer. The ladder thresholds a statistic that the unconstrained allocation drives to a hard zero, and that statistic is exactly zero in 210 of 210 surveyed (player, live half-suit) pairs. Four different ladder shapes, including one with a bottom rung at 0.10 and one with no rungs at all, produce bit-identical output. The allocator is the defect; the ladder is merely inert.

4.8 The opponent model is two scalars, and one of them is decorative

FishBot v0.4 reweights its rules posterior by a policy prior with two coefficients: a weight on "this seat asked in this half-suit" and a weight on "this seat took turns but never asked here". They are identical for all five other players and never updated during a game, so a deliberately silent opponent is misread by construction — which is what the project owner exploited at the table.

Three measured findings qualify that story, and two of them contradict the hypothesis this study began with.

1. **Deleting the prior is worse, not better.** The policy-agnostic configuration loses to the shipped one by 4.4 points [1.6, 7.2] in a paired ablation over a panel of three deceptive archetypes and the honest mirror, and by 4.60 points [2.63, 6.58] against the three deceptive archetypes alone, which is the figure a claim about deception should quote. Per cell the effect is not uniform at the resolution these banks have: over 14 (seed, archetype) cells the ablation is worse in 12, tied in 1 and reversed in 1 (sign test $p \approx 0.003$). The exposure is over-weighting, not weighting: doubling the pair costs 5.3 points.
2. **The silence channel does not exist.** Deleting the second coefficient entirely costs nothing measurable anywhere — 49.00% [45.8, 52.2] against the honest mirror, a paired delta of +0.3 points over a four-opponent panel, and a fifteen-fold sweep that is flat inside its intervals. The algebra explains it: the exponent rearranges so that the second term is card-independent, and the capacity normalisation then removes it. The channel the owner's manoeuvre attacks is carrying no information to begin with.
3. **The damage deception does is a signed, cell-local miscalibration.** Against a table of feinting opponents, a seat that has asked once in a half-suit really holds a given unresolved card of it 28.2% of the time and the policy believes 37.8% — an over-confidence of +0.096, rising at two asks. Against honest opponents the same cells are calibrated to within 0.07 and conservative. The exploit runs through the hard certificate, which the soft prior only amplifies; global belief accuracy is essentially untouched, mean absolute marginal error moving from 0.2255 under an honest opponent to within ± 0.004 of it under every archetype tested.

No archetype converted its corruption of the policy's beliefs into wins: shipped FishBot v0.4 beats all three (54.0%, 56.1%, 67.8%). The silent figure here is against the hit-probability-capped variant of that archetype, which is the competent one and the one this subsection's belief measurements use; §7.6 runs the uncapped variant, on which both policies score far higher, so the two figures are not the same measurement. The owner's live result is therefore not reproduced by commitment deception alone, and §9 records what distinguishes the two settings.

4.9 The value function carries one feature, and the mirror was never fitted against

The sixteen-feature value model is, to measurement precision, one feature. Its held-out game-clustered R^2 is 0.2327; the R^2 of a bias term plus the score differential alone is 0.2354, which is higher. Two of the sixteen features are algebraically the same feature, and five more cancel identically at every call site — moving all five to ± 9 changes none of 31,788 declaration decisions. Capacity is not the binding constraint: a depth-3 gradient-boosted tree on the same rows reaches 0.388. And the model is least informative exactly where it is needed most, reaching only 0.168 in the tail of games past 160 events.

Splitting the model by use is the actionable result. Deleting the ask-side expectimax entirely is worth +0.006 [-0.016, +0.029] — nothing — while reverting declarations to fixed thresholds costs +0.025 [+0.004, +0.047]. The part of the value model that earns its keep is the declaration rule; the ask-side use, which is the side the deadlock lives on, does not.

Finally, the fitting objective could not have caught any of this. The mirror was absent from the fitting panel in every recorded round, and the recorded per-opponent win rates never fall below 0.664, so no roughly even opponent was ever present. The objective is a soft minimum at $\beta = 10$, which over a panel whose win rates span about six points gives a maximum-to-minimum gradient weight ratio of 1.9: it is a weighted mean wearing the name of a minimum. That is the direct explanation for why a mirror pathology of this size survived fitting.

4.10 Channels the policy cannot express

Two capabilities are absent rather than mispriced.

The target dimension. Over 154,318 mirror ask decisions, at 46.6% of decisions at least two opponents are hard-indistinguishable holders of the card actually asked for, carrying a mean of 0.639 free bits per ask at a mean hit-probability spread of 0.081; against FishBot v0.3 the figures are 66.2% and 0.919 bits. The dominant refinement term in the ask score discards the target's identity outright, and every target-dependent quantity anywhere in the score is material — hit probability, the target's reply threat, hand size and exposure. Nothing prices what the choice of target tells a teammate. A separate verification qualifies the "free" in that description: the indistinguishability classes are defined by hard consistency, and choosing within one is not free once the capacity-marginal probabilities are compared, so the channel must be priced. Turn routing is available on the same dimension — at 45.5% of mirror decisions the actor could hand the turn to any of two or more chosen opponents by a provably dead ask — and the policy has one soft penalty for donating the turn to a dangerous seat and no concept of choosing whom to donate it to.

A knowledge model of anybody else. The policy builds deduction state only for itself; the only such objects it ever constructs are clones of its own. Every technique that turns on what another seat knows — teammate-directed signalling, blackballing, choosing the best-informed declarer, baiting — is out of reach not because a feature is missing but because the state it would read does not exist. Since the transcript is public, this is nearly free to add: seat j's knowledge is the public deduction state plus a posterior over j's hand.

4.11 Summary of confirmed defects

5 The v0.5 mechanisms

FishBot v0.5 is a new policy registered alongside FishBot v0.4, which stays unchanged and remains the reference opponent. Every mechanism is individually switchable, so each can be ablated; the switch

Table 3: Confirmed defects, each with the measurement that establishes it and the mechanism of §5 that addresses it. Rows marked "*designed*" are specified but not built.

Defect	**Measured cost**	**Fix**
Ownership features not gated by hit probability	39.04% dead asks, 99.3% of them voluntary; longest dead run 286	M1
Unconstrained per-card allocation guess	100% of 281 distinct forced declarations wrong; willingness ladder inert at 210 of 210 pairs	M2
Escalation stage 2 cashes without inspecting probability	8.13 points against a mirror opponent, zero against every weaker style	M8
Negative value-of-information term	weight -2.6534 on outcome entropy; most informative ask ranks 17.6 of $\approx$40	M3 (*designed*)
Refinement term discards the target's identity	0.639 free bits per ask at 46.6% of decisions	M5 (*designed*)
No knowledge model of other seats	every "what does he know" technique unreachable	M4 (*designed*)
Deterministic argmax ask selection	leakage pinned at the deterministic bound [15]; 40.03% exact repeats	M6 (*designed*)
Opponent model is two fixed global scalars	silence channel measurably inert; feint miscalibration +0.096	M7 (*designed*)
Value model is effectively one feature	held-out R^2 0.2327 against 0.2354 for bias plus score differential	M9 (*designed*)
Mirror absent from the fitting panel	soft-minimum gradient ratio 1.9 at $\beta = 10$	M10 (*designed*)

names are given with each mechanism. Three are built and measured (§5.1); seven are specified and not built (§5.2); four plausible repairs were measured, rejected, and are recorded here so that they are not rebuilt (§5.4).

5.1 Built: M1, M2 and M8

M1 — live-ask gating and ownership scaling (`m1`, `m1p`). The candidate set is restricted to asks that are not provably dead in the sense of Definition 1, computed from the actor's own deduction state. Because the predicate is a hard deduction rather than a posterior quantity, the filter cannot introduce a modelling error; it removes moves whose probability of success is exactly zero and nothing else. When the filtered set is empty — a genuinely starved turn, 0.2% of asks — the policy falls back to the full legal set, because the rules still oblige it to move. A second half was designed alongside it and is *not* in the shipped configuration: under `m1p`, all four ownership features of §4.5 — own-set progress, team control, lock completion and the completion bonus — are multiplied by the hit probability, so that a card nobody can hand over cannot outscore a live one inside the surviving set. The two halves are separately switchable because they are separate claims: the gate changes which asks are available, the scaling changes how the survivors are ranked. The gate ships on (`liveAskGate`); the scaling ships *off* (`ownershipByP`, false in `engine/src/v05.hpp`), because it was measured and rejected — once the gate has removed the $p = 0$ case there is no dead ask left for the scaling to suppress, and what remains costs 5.6 points on top of the other mechanisms at design time and -1.33 at the frozen configuration (§7.7). Every claim in this report about FishBot v0.5's play is a claim about the gate alone.

Table 4: Single-mechanism win rate against FishBot v0.4, 200 deals in six rotations, seed 90210. The control is FishBot v0.5 with every mechanism disabled, which reproduces FishBot v0.4. These are single-opponent measurements and are not a strength claim for the assembled policy; see §7.

Configuration	Mechanism	win rate
control (all off)	—	50.17%
`m1`	live-ask gate only	49.50%
`m1p`	ownership features scaled by p	50.75%
`m2`	capacity-feasible allocation	51.08%
`stage2=0`	escalation stage 2 removed	55.17%
`norepeat`	repetition guard only	50.17%

M2 — capacity-feasible allocation (`m2`). The per-card argmax of §4.7 is replaced by a search over allocations that respect the declarer's own capacity constraints. A half-suit has six cards and a team has three seats, so the assignment space is at most $3^6 = 729$; the implementation enumerates it, rejects any assignment that places a card at a seat excluded by a certificate or that gives a teammate more cards than their remaining hand can hold, and scores the survivors with the same joint estimator the declaration rule already uses — which runs on the deployed approximate belief, so the resulting probability is an estimate and not an exact posterior value. The search is exhaustive over the feasible set and is cheaper than the dynamic program it replaces. The same routine feeds three call sites: the forced-endgame declaration, the willingness ladder that selects the forced declarer — which recovers its function for free, since the statistic it thresholds is no longer identically zero — and, under a separate switch, the voluntary declaration path.

M8 — termination without misdeclaration (`stage2`, `norepeat`). The second stage of the event-count escalation is off by default; the first stage, which requires a better-than-even allocation probability, is the only escalation that remains. With M1 in place the deadlock that the second stage existed to break is gone. The design intended to put a structural backstop where the event-count guillotine had been — a repetition guard preventing the same agent from asking the same (card, target) pair twice, applied inside the candidate filter and relaxed before the starved-turn fallback — and that guard is implemented, switchable, and *off* in the shipped configuration (`repeatGuard`, false in `engine/src/v05.hpp`): it costs 6.0 points against FishBot v0.4 at design time and -5.87 at the frozen configuration, for the reason §7.7 gives. M8 as shipped is therefore a deletion and nothing else. The design rule behind it is the project owner's, and it is a claim about the game: a risky ask is better than a declaration the policy itself scores at zero.

What each is worth alone. Each mechanism was measured in isolation against FishBot v0.4 over 200 deals in six seat rotations at seed 90210, with FishBot v0.5 configured to reproduce FishBot v0.4 exactly as the control (50.17%). The results are in Table 4. Two readings are worth stating plainly. First, the three mechanisms that remove the pathology are close to strength-neutral in head-to-head play, and the one that is not — M8 — gains by *withholding* a bad declaration rather than by playing better. Second, the repetition guard on its own is exactly neutral, which is consistent with §4.5: with the dead asks already gone, there is almost nothing left for it to catch.

The pathology gates. The build order was gated on the three pathology statistics rather than on win rate, following the standing preference that robustness is the research question. §7 reports the full comparison; the gates themselves were: after M1, no game may contain a dead run of six; after

M2, forced-endgame accuracy must leave zero; after M8, no game may reach the action cap and no declaration may be made past the horizon.

The parameters behind this table are not FishBot v0.5's. Table 4 was measured before the refit, with the 34 fitted coordinates still FishBot v0.4's, and on the design-time assembly, which carried both of the mechanisms this subsection has just recorded as rejected. They had been fitted with the escalation of §4.6 present as a safety net, so with that net removed that assembly under-claims: it makes 4.21 declarations per game against FishBot v0.4's 4.75, and scores 41.8% [39.25–44.50] head-to-head. Most of that shortfall belongs to the two rejected mechanisms rather than to the coordinates, and the distinction matters because the figure has been read as a parameter mismatch: on the same bank, with the same FishBot v0.4 coordinates, the assembly that actually ships scores 49.08% [46.67–51.58], which is 7.3 points higher and level with FishBot v0.4 (`research/v05/runs/v04vector-in-v05.txt`). §6 describes the refit that replaced those coordinates in the frozen configuration, and every strength figure for FishBot v0.5 in this report is measured after it; the ablations of §7.7 repeat this table's comparisons on the fitted vector, and its control is correspondingly not FishBot v0.4.

5.2 Designed and not built

None of the following is on FishBot v0.5's decision path. Prototype code exists outside it for the target-scoring and opponent-model mechanisms, and is used only to check that the quantities they need can be computed; nothing below has been measured in play, and each entry states the constraint the diagnosis places on its eventual construction.

M3 — a net-information term in the ask score. Replace the negative value-of-information term with a signed net term: the information the ask delivers to teammates, minus the information it delivers to opponents, weighted separately. This is the secrecy-rate trade-off of §3.3, and the two filters the engine already produces — the policy-agnostic posterior and the policy-weighted one — are enough to compute both sides. The design constraint established in §4.4 is that the channel is almost never *provably* free: it must be priced against the probability that the team owns the half-suit, which at the relevant states averages 0.068 and peaks at 0.342, not gated on a certainty no observer has. A FishBot v0.5 that signals only when signalling is free will never signal. M3 requires M4.

M4 — a knowledge model of the other five seats. Maintain, for each seat, the public deduction state plus a posterior over that seat's hand. Nearly free because the transcript is public, and a precondition for M3, M5 and M7.

M5 — target-dimension selection. Score the target explicitly: lockout value (which opponent must not receive the turn), void progress inside the half-suit, and, where conventions are enabled, codebook value. The measured headroom is in §4.10, and the verification recorded there requires that the choice be priced against the capacity-marginal hit probability rather than treated as free.

M6 — partner-aware stochastic action selection. Select by softmax over the near-optimal set rather than by argmax. This is required for any leak penalty to mean anything at all: leakage is a function of the distribution over actions, so a deterministic policy sits at the deterministic bound whatever its feature weights [15]. The temperature and the convention flag are both keyed on the partner regime. With bot teammates — the self-play condition, in which partners follow a known blueprint — lower temperature and convention-rich play are appropriate, because the receiver decodes what the sender encodes. With human or unknown teammates the blueprint assumption fails, and play must be grounded and less predictable, because an opponent who has watched a deterministic argmax across many games learns to read it. The two regimes are to be reported separately, and the self-play configuration is never to be headlined as though it were the one a human will meet.

M7 — an online per-seat opponent model. Replace the two global scalars with a per-player posterior over a small type library, updated from that player's public asks and from a *time-local* silence statistic — own turns since last asking in this half-suit, and having been asked in it without replying — rather than from a whole-game ask count, which §4.8 shows is inert. Shape it as a data-biased response [22]: confidence per information set, decaying to the population prior where evidence is absent, so that a deliberately silent opponent produces no evidence and receives the prior instead of being misread. Expose the exploitation/robustness dial explicitly and report the frontier [21] rather than a single operating point.

M9 — the value model, rebuilt or retired. Two options, to be decided by measurement: collect rows at declaration points as well as ask points, drop the degenerate features, and refit jointly with the policy vector so that the freeze step covers it; or retire it and score declarations directly on the allocation probability against a fitted threshold. §4.9 shows that the ask-side use contributes nothing and the declaration-side use does, and that a tree model finds real structure the linear form cannot express.

M10 — fitting and evaluation. Put the mirror in the panel. Replace the soft minimum, which at the panel's spread is a weighted mean (§4.9), with a worst-case or minimax-regret objective over the style set, and report both the minimum and the regret — a raw minimum over a small style set stagnates on the hardest member [29]. Promote the three pathology statistics — longest dead run, share of games with a dead run of six, share of post-horizon declarations that are wrong — to first-class metrics gating every commit.

5.3 Conventions, and which configuration is the headline

The strategy corpus states that pre-agreed conventions are outside this game [4]; the rules corpus documents a named convention in real use [3]; and a pair of policies trained together has an implicit pre-agreement whether or not anyone writes it down. FishBot v0.5 therefore builds the signalling machinery behind an explicit switch, ships it *off* as the headline configuration — which is legal under every reading of the rules we found — and publishes the with/without difference as a result in its own right rather than folding it into a single number.

5.4 Measured, rejected, and recorded so as not to be rebuilt

1. **A time-varying holding cost in the value model.** The literature prescription of §3.2 — make waiting strictly costly and the stopping problem becomes well posed [11] — was implemented as a linear penalty in elapsed events and swept over a forty-fold range of magnitudes. It is a mean-shifter. Events per game fall from 146 to 121 and the frequency of dead runs from 32% to 18.5%, but the tail does not move at all: the 90th percentile of events goes from 313 to 311 and the longest dead run from 289 to 289. Declaration error rises from 10.9% to 13.4% and the configuration loses 3.9 points head-to-head. A constant holding cost is also not missing — the declaration margin already is one — and in deadlocked positions the urgency flag bypasses the value-based declaration rule entirely, so the term binds in fewer than one in ten late blocked opportunities.

2. **Deleting the policy prior.** Worse, not better: by 4.4 points [1.6, 7.2] on a mixed panel of three deceptive archetypes and the honest mirror, and by 4.60 points [2.63, 6.58] against the deceptive archetypes alone (§4.8). The exposure is over-weighting.
3. **A turn-transfer willingness ladder.** A genuine multi-candidate transfer decision arises 0.148 times per game and governs 0.92% of asks; the unilateral choice already matches an omniscient oracle at 800 of 886 of them, and handing one team a ground-truth chooser is worth -0.0007 ± 0.0024 sets per game over 6,000 paired games — indistinguishable from zero. The channel is also the leakier of the two: an eight-rung ladder transmits 2.19 bits per candidate against the one bit the rules license, and it transmits them mid-game to live opponents. Budget nothing for it.
4. **Confidence-ranked declaration arbitration.** Resolving simultaneous declarations by lowest seat rather than by confidence costs 0.37 percentage points pooled over 30,000 games. The rules-legal willingness ladder may be added if it is convenient, but nothing should be engineered for it.

6 Fitting

The mechanisms of §5.1 change what the ask score has to discriminate between, and they remove a forcing rule that the declaration thresholds had been fitted underneath, so the compiled coordinates were refitted. This section describes that refit, and is deliberately explicit about how little of it there is: one round, one base seed, and a value model that was not touched.

It should be read alongside a measurement of what it was worth, because the refit is not what produces the result of this report, and the length of this section is not evidence that it is. Carrying FishBot v0.4's frozen vector unchanged, the shipped mechanisms play level against FishBot v0.4 — 50.22% [48.40–52.03] over 6,000 games at 2 banks, range 50.13–50.30%, and 4.49 declarations per game against 4.47 — which is the same wash §7 reports for the fitted vector. In the mirror, on the instrument and seed of §7's pathology bank, that same unfitted combination already takes the dead-ask rate to 0.02%, the longest dead run to 1, games containing a run of six or more to 0%, exact repeats to 2.97%, declarations at or after the old horizon to 0, and declaration error to 1.89% (`research/v05/runs/v04vector-in-v05.txt`). On every row of that instrument the repair is therefore mechanical rather than parametric — the one exception being the forced endgame, which fires 4 times on a bank this size and so cannot separate the two at all. The pre-refit shortfall reported in §5.1, 41.8%, is not evidence against this: it was measured on the design-time assembly, and 7.3 of its points belong to two mechanisms that were subsequently rejected and do not ship.

What the refit does, then, is re-seat 14 decision knobs that were fitted underneath a forcing rule which no longer fires. Every figure in §7 is measured at the vector it produced and is conditional on that vector; none of them is attributable to the refit rather than to the mechanisms, and this section should not be read as though they were.

6.1 The parameter vector, and why its offset is derived

The fitted vector has 34 coordinates, in the same layout as FishBot v0.4's so that a FishBot v0.4 vector can seed a FishBot v0.5 fit: the 20 ask-score weights $w_0, \ldots, w_{19}$, followed by 14 decision knobs — the two declaration thresholds, the ask floor, the patience pool, the opponent-card floor, the two mixing weights, the minimum team probability, the stopping margin, the two policy-prior coefficients θ and ϕ, the lookahead width, and the chain and threat weights. Each knob is clamped to its own interval as the vector is parsed, so that probabilities stay in $[0, 1]$ and counts stay non-negative integers.

The value model is *not* in this vector, exactly as in FishBot v0.4. Mechanism M9 of §5.2 left open whether to refit it jointly inside the search or retire it in favour of scoring declarations directly on allocation probability, and neither was done: the 34 policy coordinates were fitted conditional on sixteen value coefficients that were never re-estimated, and whose measured behaviour §4.9 describes.

The one structural change is the offset at which the decision knobs begin. It is now *derived* from the compiled ask-feature count in both places that touch the flat vector — the spec parser in `engine/src/factory.hpp` and the freezer in `engine/freeze_config_v05.py` — rather than written down in either. The v0.4 pipeline hard-coded the offset at 18. When the ask-feature count later grew to 20, the last two ask weights were read as the first two decision knobs, and every knob after them shifted: the configuration the optimiser scored was not the configuration the freezer baked in. Nothing in the pipeline could notice. Both halves ran without error, and both produced a policy that plays legal Fish.

FishBot v0.5 therefore carries a mechanical assertion in place of a convention. The shipped configuration is played against the flat vector that is supposed to encode it: `v05` on one side, and on the other `v05:allparams=` followed by the frozen vector. Two identical policies meeting on deals each played in both team orientations must split every deal, so the assertion is that the match returns exactly one half, and any other value is a mismatch between the two representations. It passes: 50.0000% over 6,000 games at 2 independent seeds, with a deal-clustered interval that is a single point (`research/v05/runs/roundtrip-assert.txt`).

An assertion that always passes is worth showing to have teeth. Re-encoding the same vector so that a correct parser reconstructs exactly what a parser at offset 18 would have built produces a policy that loses to the shipped one: it scores 45.87% [44.07–47.70] over 3,000 games, short of even by 4.13 points. That is the magnitude of the silent error the v0.4 pipeline could not detect, and it is larger than most of the differences this report goes on to measure.

One caveat belongs beside the assertion rather than in a footnote. The freezer writes ask weights to four decimal places and knobs to five, so what ships is a rounding of the optimiser's output and not the output itself. Played against the baked configuration, the full-precision vector scores 50.43% [49.43–51.40] over 3,000 games — no measurable difference, but not an identical policy — so the round trip is asserted, and run, at the precision that actually ships. The freezer's own convenience output does not currently do that: it prints a suggested assertion command with every coordinate formatted at five decimals, which therefore does not return one half. The vector it writes is correct and the assertion above is run against that; the printed command is recorded as a defect to fix, because a provenance check that fails silently is worse than none, and that is the class of defect this whole subsection exists to prevent.

6.2 Optimiser and schedule

The optimiser is unchanged from FishBot v0.4: the cross-entropy method with a diagonal Gaussian over the 34 coordinates (`engine/src/tuner.hpp`). Each generation samples 24 candidates around the incumbent mean, always evaluating the incumbent itself as one of them, scores every candidate against every panel member, keeps the 6 best as the elite set, and smooths the mean and the per-coordinate standard deviation towards the elite statistics. Every candidate within a generation is evaluated on *identical* seed banks, so within-generation comparisons are paired and the deal draw cancels; the banks are redrawn between generations, so the search cannot settle onto one set of deals.

The run was 40 generations at 200 deals per panel member per candidate, each deal played in both team orientations (400 games per cell, 2,304,000 games in total), from base seed 505101. The initial mean was FishBot v0.4's frozen configuration, so what follows is a local refinement of FishBot v0.4 and not a search from scratch. The base seed is recorded in §9 of `docs/FISHBOT_V05.md` and in the artifact manifest; that closes one v0.4 gap cheaply, since the round that produced FishBot v0.4's shipping configuration has no committed script carrying its base seed and can be inspected but not reproduced.

6.3 The objective, and the two corrections it makes

The quantity optimised is the same soft minimum over an opponent panel $\mathcal{O}$ that FishBot v0.4 used. For a parameter vector w, writing $\mathrm{wr}_o(w)$ for its win rate against opponent o on the current bank,

$$\mathrm{score}(w) \;=\; -\frac{1}{\beta}\,\log \sum_{o\in\mathcal{O}} \exp\!\big(-\beta\,\mathrm{wr}_o(w)\big). \tag{1}$$

Two things about it changed, and §8 records both as corrections to the v0.4 study rather than as improvements on it.

The panel now contains an opponent of the agent's own strength. It is v0.4, v0.3, lockout, detective, diversifier, hunter — 6 members, with FishBot v0.4 at the head. No recorded round of the v0.4 fit contained a self-play or mirror-strength opponent at any point. §4.9 gives the recorded per-opponent floor over every committed v0.4 trace; a policy of a candidate's own strength sits near one half by construction, which is well beneath it. A candidate that deadlocked against a copy of itself while converting against everything weaker therefore scored exactly as well as one that did not. The pathology of §4 was not traded away during fitting; it was never priced.

β is 25 rather than 10. A soft minimum is only a minimum if its gradient weights separate. The weight that (1) places on opponent o is proportional to $\exp(-\beta\,\mathrm{wr}_o)$, so the ratio between the largest and the smallest weight on a panel is exactly $\exp(\beta\Delta)$, where Δ is the spread of win rates across that panel. FishBot v0.4's selected profile spans $\Delta = 0.063$ over its 4 opponents, and at $\beta = 10$ that is a ratio of 1.9. Weights lying within a factor of two of one another describe a weighted mean, not a minimum, which is why the v0.4 description of the objective as a worst case across playstyles is corrected rather than repeated. FishBot v0.5's panel profile spans $\Delta = 0.415$, and at $\beta = 25$ the ratio is 32048.3.

Neither change alone accounts for that. The same $\beta = 25$ applied to FishBot v0.4's narrow spread would give a ratio of only 4.83, and $\beta = 10$ applied to FishBot v0.5's spread would give 63.43. The temperature sharpens the objective; putting an even-strength opponent in the panel is what gives it something to sharpen.

What is still missing is the objective M10 actually asks for. This remains a soft minimum with a raised temperature, not an explicit worst case, and not the minimax-regret objective of §5.2. Regret over the style set is reported in §7; it was never optimised, and FishBot v0.5 does not come out ahead of FishBot v0.4 on it.

6.4 The winner's-curse guard, and how to read a per-generation number

The score a cross-entropy trace reports for a generation is the best of 24 candidates evaluated on one shared seed bank. It is a maximum over a noisy sample, so it estimates the winner's luck along with its strength. The tuner guards against shipping that luck: after the final generation it re-evaluates two candidates — the distribution mean, and the single best candidate seen in any generation — at one common seed that neither was selected on, and returns whichever of the two scores higher.

Here it returned the mean, which scored 0.5272 against the best generation's 0.4798, a gap of 0.0474. The distribution mean is what `engine/freeze_config_v05.py` baked into `V05Config`. The number to hold next to that is what the losing candidate had scored in its own generation: 0.5693 at generation 24, on a panel profile of 57.0 / 75.2 / 80.5 / 79.0 / 94.8 / 98.5. The same vector, re-evaluated on a bank it had not been selected on, scored 0.4798.

That single comparison is the reading rule for the whole trace, and it is worth stating as a rule because per-generation numbers are the ones a reader is most likely to quote. The per-generation scores in `research/v05/runs/fit-round1.jsonl` are *selection statistics*: they rank candidates within a generation against a common bank, which is what they exist for, and they are not measurements of a candidate's strength. The best one in this trace overstated its own vector by 0.0895 on the objective's scale. Only the final common-seed re-evaluation, and the held-out banks of §7, are measurements.

6.5 The held-out regression

The best generation's own profile puts FishBot v0.5 at 57.00% against the mirror-strength member of the panel. Every seed bank used from §7 onwards is disjoint from the fitting banks, and the configuration was frozen before any of those runs. On them the same configuration wins 50.77% of games against FishBot v0.4 across 5 seeds (range 50.06–51.39%, 9,000 games), and 51.61% [49.28–53.94] on the per-style bank. In-panel 57.00% against held-out 50.77% is a regression of 6.23 points, and the held-out figure is the one this report uses everywhere the two could be confused.

The regression is what the guard above predicts and does not remove. The guard protects the *choice* between two candidates from selection bias; it does nothing about the bias in the in-panel figure itself, which is a maximum taken over 24 candidates on a shared bank in each of 40 generations. An in-panel number from this pipeline should be read as an upper estimate by construction, and the size of the gap here — rather than any argument about it — is the calibration.

One further scope limit belongs here rather than in §9. Deals were held out; opponent *identities* were not. All 6 of the fitting panel's members reappear in the evaluation panel of §7, so the held-out figures measure generalisation to new deals rather than to new strategic populations. The three deceptive archetypes of the deception panel are the exception: none of them was in the fitting panel, and FishBot v0.5's loss against one of them is reported there.

6.6 What this fit is not

- **It is one round.** FishBot v0.4 ran 5 rounds; this is 1, from a single base seed, initialised at FishBot v0.4's frozen vector. Nothing here distinguishes a converged optimum from the nearest local one, and no second round from a different seed was run to check that the search lands in the same place twice. Every strength figure in §7 is conditional on that.
- **The value model was left where it was.** It was fitted separately for FishBot v0.4 by ridge regression on self-play decision points and then held fixed while the policy weights moved around it, and that arrangement is unchanged. §4.9 measures what the model contributes and where; the refit did not act on it.
- **The panel is still unbalanced.** As the profile in §6.4 shows, most of its members are dominated by a wide margin, and a dominated opponent contributes almost no gradient at any temperature. The resolution the objective has is spent largely where the outcome was not in question.
- **One fitted coordinate moved into a known exposure.** The policy prior's θ, the weight on "this player asked in this half-suit," rose from 0.264 to 0.445. The diagnosis had already established that over-weighting that prior is precisely the deception exposure — while deleting it outright is worse still against the deceptive archetypes themselves, by 4.60 points [2.63, 6.58] — and the deception panel of §7 measures the bill: -2.2 points against the archetype that manufactures a false ask-legality certificate. The objective could not have seen this, because no deceptive archetype was in the fitting panel either.

7 Results

The result of this work is the removal of the failure mode of §4.2, and it is not a strength jump. Over the 9 opponents of Table 8 the two policies' mean win rates are 83.65% and 83.57%, their worst cases

Table 5: `fish pathology` run on the FishBot v0.4 mirror, the FishBot v0.5 mirror and the cross match. Population: 300 deals per column at both team orientations; seed 31 for the two mirrors and 90210 for the cross match, full rules dialect, both policies frozen. The cross match is 600 distinct games. The two mirror columns are 300: in a mirror the two orientations run identical policies and play byte-identical games, so the instrument's raw counts are each game counted twice, and every count in those two columns is reported here per distinct game — the same halving C3 in §8 applies to the forced-endgame figure. Re-running each bank at one orientation returns exactly half of every count and every rate to the printed digit, so no rate in this table is affected. The instrument also pools *both* teams, which is why the cross-match column is a property of the pair and not of either policy: the dead asks it counts are FishBot v0.4's, and §7.3 returns to the consequence of that pooling for the forced-endgame row. No intervals; these are counts and rates over one bank per column. Source: `research/v05/results/E2-pathology.txt`.

	FishBot v0.4 mirror	**FishBot v0.5 mirror**	**cross**
events per game	143.6	96.7	100.0
median / p90 / p99	106 / 312 / 321	96 / 112 / 124	99 / 116 / 134
asks per game	134.3	87.4	90.7
ask hit rate	34.25%	55.63%	54.59%
provably dead asks	39.04%	0.01%	3.12%
dead runs	1,305	3	1,696
longest	286	1	1
games with a dead run ≥ 6	34.33%	0%	0%
starved turns	0.2%	0.01%	0.0018%
exact repeat (actor, card, target)	40.03%	2.61%	5.04%
asks inside a half-suit the team owns	16.46%	9.75%	9.18%
declarations	2,700	2,700	5,400
wrong	10.44%	2.07%	2.2%
at or after the forcing horizon	384	0	0
of those, wrong	58.59%	—	—
forced-endgame declarations (pooled)	14	3	26
of those, wrong	100%	100%	88.46%
games reaching the action cap	0%	0%	0%

are 51.61% and 50.00%, and FishBot v0.4 is marginally the better of the two on minimax regret over that style set. The evidence for FishBot v0.5 is Table 5, the forced-endgame measurement of §7.3, and the gain of +7.2 points against the deception archetype that motivated the work, replicated at both of the banks in Table 9. The same table records a smaller and equally replicated *loss* of −2.2 points against a different deceptive archetype, and the evidence against a strength claim is that loss together with every row of Table 7 and of Table 8. All of it is reported below at the same weight.

7.1 The pathology, on the same instrument

Table 5 is the paper's result. The three statistics promoted to commit gates in §5.1 — longest dead run, share of games containing a dead run of six, and declarations made at or after the forcing horizon — read 1, 0% and 0 on the FishBot v0.5 mirror bank, against 286, 34.33% and 384 for FishBot v0.4 on the identical bank. The provably-dead ask rate falls from 39.04% to 0.01%, which is a property of the gate rather than a measurement: the predicate of Definition 1 is the same deduction the filter runs, so a gated policy can record a dead ask only through the starved-turn fallback, and on this bank the fallback never fired at all (0.01% of asks).

What is a measurement, and what the gate did not guarantee, is the rest of the table. Ask hit rate rises from 34.25% to 55.63%, and ask volume falls from 134.3 to 87.4 per game: removing a class of moves that could not succeed did not merely shorten the games, it changed what the surviving asks are spent on. The event distribution moves at the tail and not only at the mean, which is the property

Table 6: Forced-endgame declarations, per declaring team (E8). Population: 4,000 deals in six seat rotations, 24,000 games per arm, seed 909090, FishBot v0.5 against FishBot v0.4, both frozen. The two arms are the two sides of the same matches, scored separately; "rate" is forced declarations per game by that team and "correct" is the share of them that named the true allocation. Declaration totals are all declarations by that team, forced and voluntary. No intervals: a forced declaration is a rare event and these are pooled counts. Source: `research/v05/results/E8-forced-endgame.txt`.

	forced declarations		**all declarations**	
	rate/game	**correct**	**rate/game**	**correct**
FishBot v0.4	0.0303	0.14%	4.47	98.51%
FishBot v0.5	0.0048	25.00%	4.49	98.09%

the swept holding cost of §5.4 failed to obtain — the 99th percentile falls from 321 events to 124, and the median from 106 to 96. Declaration error falls from 10.44% to 2.07%, and the 384 post-horizon declarations that were 58.59% wrong have no counterpart because no game reaches the horizon.

Three rows are worth reading against the natural expectation rather than for it. Exact repeats fall to 2.61% but not to zero, and should not: cards move, so re-asking a (card, target) pair can be the best move on the board once a public transfer has put the card where it was not before. That is the same observation that sank the repetition guard (§7.7). Asks inside a half-suit the actor's own team already owns fall from 16.46% to 9.75%, and again not to zero: those asks are legal, are not provably dead when the actor cannot establish the ownership, and emit ask-legality certificates. FishBot v0.5 does not choose them for that reason — the net-information term M3 is unbuilt — so the remaining rate is incidental, exactly as §9 says of the FishBot v0.4 rate. Finally, the cross-match column records 3.12% dead asks with every dead run of length 1: the instrument pools both teams, these asks are FishBot v0.4's, and the run lengths are one because the seat that would have continued the cycle is now gated.

7.2 Engine and information-safety verification

`fish verify` plays the ordered cross-product of the engine's 9-policy pool — 81 ordered pairs over a budget of 600 games — with the knowledge audit active, and records 0 violations in 23,594,580 soundness checks: no derived knowledge state, capacity or certificate was ever contradicted by the actual deal. Card conservation holds in every game, with 0 set-conservation failures; 0 games reach the action limit; and bit-identical replay is PASS.

The scope of that sweep needs stating, because it is narrower than it looks. The pool it plays is FishBot v0.4, FishBot v0.3, FishBot v0.2 and the six scripted styles, and it does *not* contain FishBot v0.5; the determinism check is run on a FishBot v0.4-against-FishBot v0.3 bank. What E1 establishes is that the rules driver, the public deduction state and the certificate machinery are sound — which is the machinery M1's gate is a pure function of, so a sound deduction state is exactly the precondition the gate needs — and not that FishBot v0.5's own play was audited. For FishBot v0.5 the corresponding evidence is the action-limit rate, which is 0% for the FishBot v0.5 mirror and 0% for FishBot v0.4 on the bank of Table 5, and is non-zero on 0 of the 37 match rows behind the tables in this section: no result here is an artefact of the cap ending a game. Sources: `research/v05/results/E1-verify.txt`, `engine/src/main.cpp`.

7.3 The forced endgame, at volume

The forced endgame is too rare for the pathology bank of Table 5 to resolve. Its FishBot v0.5 mirror column counts 3 of these declarations in 300 games, which is a denominator that supports no rate. E8 therefore plays 24,000 games per arm and scores each team separately. Table 6 gives the result.

Table 7: FishBot v0.5 against FishBot v0.4, five independent evaluation banks (E3). Population: 300 deals in six seat rotations per bank, 1,800 games per row and 9,000 in total, full rules dialect, both policies frozen. Intervals are deal-clustered percentile bootstrap. The deals are held out from the fit of §6; FishBot v0.4 itself was a member of the fitting panel, so the opponent identity is not. Source: `research/v05/results/E3-headtohead.jsonl`.

Bank seed	FishBot v0.5 win rate	95% CI
90210	50.50%	48.33–52.67%
31337	50.06%	47.67–52.50%
515151	51.39%	48.89–53.83%
777001	51.22%	49.00–53.44%
424242	50.67%	48.28–53.06%
mean of the five	50.77%	range 50.06–51.39%

FishBot v0.5 enters the forced endgame 6.3× less often — 0.0048 declarations per game against 0.0303 — and when it does, it names a capacity-feasible allocation that is correct 25.00% of the time (29 of 116 forced declarations), against 0.14% for FishBot v0.4 (1 of 728). The denominators are worth carrying: the FishBot v0.5 rate rests on 116 events, which is enough to separate it from FishBot v0.4's and not enough to place it to the nearest point. E8 does not decompose the two effects by mechanism, so the attribution that follows is the design's and not this experiment's: M1 is what removes the extended runs of guaranteed misses during which a team is stripped of cards while half-suits remain undeclared, which is how the cardless endgame was being reached, and M2 is what stops the declarer, once there, naming an allocation that gives a teammate more cards than that teammate can hold.

25.00% is not the ceiling. §4.7 measures the best capacity-feasible allocation to be correct 40.6% of the time under the reference posterior. That ceiling was measured on FishBot v0.4's mirror forced-endgame states, not on the states FishBot v0.5 reaches here, so it bounds this number only to the extent that the two populations resemble each other; read as an indication rather than a bound, roughly two-fifths of the available accuracy remains unclaimed. The gap is a live target rather than a residual: M2 ranks feasible assignments by an estimate computed on the deployed approximate belief, and it does not attempt the per-seat knowledge model (M4) that would sharpen the ranking.

Why this is reported per declaring team. `fish pathology` pools both teams' forced declarations into a single count. In the cross match of Table 5 that pooled figure is 26 declarations, 88.46% wrong, and it is a property of neither policy: Table 6 scores the same kind of event separately by declaring team, on a far larger bank, and finds 25.00% correct for FishBot v0.5 against 0.14% for FishBot v0.4, so the pooled cross-match figure is dominated by FishBot v0.4's side of those games. This is the same class of error corrected at C3 in §8, where a mirror run's six seat rotations were read as six independent samples although they are three deal shifts times two team orientations and both orientations play byte-identical games. Forced-endgame accuracy is attributable only when it is reported per declaring team, which is why Table 6, and not the pathology row, is the measurement of record.

7.4 Head-to-head against FishBot v0.4, on deals never fitted against

Across the five banks of Table 7, FishBot v0.5 wins 50.77% of 9,000 games against FishBot v0.4, with individual banks spanning 50.06–51.39% and every interval containing 50%. The mean margin is about a point, which is smaller than the spread between banks, and no claim of a strength advantage is made from it.

These are not the only banks in this report on which the frozen configuration meets FishBot v0.4. Counting the per-style cell of Table 8 (51.61%, the figure the abstract and §1 quote because it is the

Table 8: Per-opponent win rate for both policies against the same 9-member style set (E4). Population: 300 deals in six seat rotations, 1,800 games per cell, seed 515253, full rules dialect, both policies frozen. Δ is FishBot v0.5 minus FishBot v0.4 in percentage points. The FishBot v0.4-against-FishBot v0.4 cell is 50% by construction, not by measurement. Minimax regret is computed against the better of the two *measured* arms on each style, not against a per-style optimum, which was not computed (§9). Source: `research/v05/results/E4-perstyle.jsonl`.

Opponent	**FishBot v0.5**	**FishBot v0.4**	**Δ**
FishBot v0.4	51.61%	50.00%	+1.61
FishBot v0.3	72.28%	73.33%	−1.06
FishBot v0.2	81.72%	83.06%	−1.33
lockout	80.00%	77.94%	+2.06
detective	75.61%	77.28%	−1.67
diversifier	94.11%	92.89%	+1.22
hunter	97.67%	97.67%	0.00
bluffer	99.89%	99.94%	−0.06
random	100.00%	100.00%	0.00
worst case	**51.61%**	**50.00%**	
attained on	v0.4	v0.4	
mean over the style set	83.65%	83.57%	
minimax regret	1.67	2.06	
attained on	detective	lockout	

cell the worst case of that table is read from), the shipped row of the ablation bank (49.40%) and the default dialect of Table 12 (50.20%), 8 independent banks measure the same pairing and they span 49.40–51.61%. That span, and not any one of its members, is what this report knows about the head-to-head: it straddles even money, its width exceeds every margin inside it, and its lowest member, 49.40%, puts FishBot v0.5 below FishBot v0.4.

The fitting regression, stated plainly. The generation the fit of §6 selected from scores 57.00% against FishBot v0.4 in-panel, on the banks it was being optimised on. Held out, after freezing, the same configuration scores 50.77%. In-panel 57.00% versus held-out 50.77% is the regression, and the held-out figure is the one reported here and everywhere else in this paper. Nothing about the in-panel number should be quoted as a result.

The composition of the margin is worth recording because it is not the composition one would predict from the pathology table. FishBot v0.5 takes 4.53 half-suits per game against 4.47, and declares slightly more often — 4.51 declarations per game against 4.45 — at a slightly *lower* accuracy, 98.18% against 98.50%. FishBot v0.4 gets 1.50% of its declarations wrong in these games, against 10.44% in its own mirror, because against a FishBot v0.5 opponent the games end at 99.9 events and FishBot v0.4 never reaches the horizon that forces its bad declarations. The failure mode of §4.2 is a property of the interaction, not of one seat's rule in isolation, and repairing one side of the match removes most of it from both.

7.5 The per-style profile, and the worst case

Table 8 is the table this project's standing evaluation preference asks for, and it says that on win rate the two policies are level. FishBot v0.5 is ahead on four styles, behind on four, and identical on one; the largest gain is +2.06 against the turn-starvation lockout and the largest loss −1.33 against FishBot v0.2. The two means differ by 0.09 points, computed from the unrounded cells. On minimax regret FishBot v0.4 is marginally the better policy, 2.06 against 1.67, its regret attained on lockout and FishBot v0.5's on detective. Neither figure supports an ordering.

Table 9: FishBot v0.5 and FishBot v0.4 against three deceptive archetypes (E10), at two independent seed banks. Population: 400 deals in six seat rotations, 2,400 games per cell, seeds 31415926 and 8675309, full rules dialect, both policies frozen. All three are commitment deceivers: they follow a fixed deceptive rule for the whole game. *Silent* never asks in the half-suit it holds most cards of until forced — the uncapped variant, not the hit-probability-capped one whose figure §4.8 quotes; *feint* preferentially asks in a half-suit it holds exactly one card of, manufacturing a misleading ask-legality certificate when the material cost is small; *withholder* is the project owner's own manoeuvre, declining to ask in a half-suit for several turns after being asked in it while holding other cards of it. Δ is the mean over both banks of FishBot v0.5 minus FishBot v0.4. Source: `research/v05/results/E10-deception.md`.

	seed 31415926		seed 8675309		
Opponent	**FishBot v0.5**	**FishBot v0.4**	**FishBot v0.5**	**FishBot v0.4**	Δ
silent	80.42%	79.96%	83.17%	79.00%	+2.3
feint	50.96%	54.13%	52.08%	53.29%	−2.2
withholder	73.63%	66.25%	71.42%	64.46%	+7.2

The worst case, and what FishBot v0.4's published one excluded. The v0.4 study reported that FishBot v0.4's lowest win rate against any panel member was 75.07%. That panel contained no opponent of FishBot v0.4's own strength. Once one is included, the worst case for both policies falls to about 50%: 51.61% for FishBot v0.5 and 50.00% for FishBot v0.4, both attained on v0.4. This is not a defect that a better policy in this family would repair. A policy's mirror match is a two-team game between identical strategies, so FishBot v0.4's own cell is exactly 50% by construction, and any policy strong enough that no member of a fixed panel beats it will have its worst case set by a copy of itself. A worst-case figure quoted over a panel that excludes the mirror is therefore a statement about the panel. C9 in §8 records the scope correction; the operational consequence is M10's, and it is that the mirror belongs in the panel that the worst case is computed over as well as in the panel that is fitted against.

7.6 The deception panel

The withholder is a direct model of the manoeuvre reported from live play that started this work: hold cards of a half-suit you have been asked for, then decline to ask back in it. Against that archetype FishBot v0.5 gains +7.2 points, and the two banks agree closely on the size of it — +7.38 and +6.96 — which is the strongest replication anywhere in this report. It also gains +2.3 against the silent archetype, but there only the sign replicates: the banks read +0.46 and +4.17, and the first of those is a fraction of the width of a bank, so the silent result should be read as a direction and not as a magnitude. The mechanism is not a better opponent model — M3 through M7 are unbuilt and the two policy-prior scalars are the same construct in both arms. It is that FishBot v0.5 no longer spends turns on asks it can prove will miss, so a misleading *absence* of asks has much less leverage over the position. The manoeuvre works by making the deceiver's holdings appear to lie elsewhere, and a policy already spending 39.04% of its asks on cards it could prove were unavailable was donating turns for that misdirection to work in.

The loss, and its cause. Against the feint FishBot v0.5 is −2.2 points worse, and that replicates at both banks too (−3.17 and −1.21). The feint manufactures a false-positive ask-legality certificate rather than withholding a true one, and the fit raised `priorTheta` — the weight on "this player asked in this half-suit" — from 0.264 to 0.445. FishBot v0.5 therefore weights a manufactured certificate more heavily than FishBot v0.4 did. This is the exposure the diagnosis predicted: §4.8 establishes that the vulnerability is over-weighting the policy prior, not weighting it at all, and the corresponding removal is worse still — deleting both prior scalars costs 4.60 points [2.63, 6.58] on a panel containing

Table 10: Frozen-parameter mechanism ablations (E5). Population: 250 deals in six seat rotations, 1,500 games per row, seed 606060, every row played against frozen FishBot v0.4, full rules dialect. Every row runs FishBot v0.5's fitted parameter vector; only the mechanism switches change, so the control is *not* FishBot v0.4 — it is the FishBot v0.5 vector with the mechanisms disabled, which is why it differs from the FishBot v0.4-parameter control of Table 4. The last two rows *add* a mechanism to the shipped configuration rather than removing one. "Decl. acc." is the ablated arm's own declaration accuracy and "opp." is FishBot v0.4's in the same games. Intervals are deal-clustered percentile bootstrap. Source: `research/v05/results/E5-ablations.jsonl`.

Switches	Mechanisms	win rate	95% CI	events	decl. acc.	opp.
`m1=0,m2=0,stage2=1`	control	50.67%	48.47–52.87%	141.8	89.10%	88.61%
`m1=1`	M1	49.87%	47.40–52.33%	99.7	98.85%	98.45%
`m2=1`	M2	52.27%	50.00–54.53%	142.0	91.02%	86.88%
`stage2=0`	M8	56.60%	54.33–58.93%	142.3	96.67%	83.55%
`m1=1,m2=1`	M1 + M2	49.40%	46.87–51.93%	99.7	98.47%	98.45%
`m1=1,stage2=0`	M1 + M8	49.87%	47.40–52.33%	99.7	98.85%	98.45%
`m2=1,stage2=0`	M2 + M8	56.33%	54.00–58.67%	142.3	96.15%	83.55%
shipped FishBot v0.5	M1 + M2 + M8	49.40%	46.87–51.93%	99.7	98.47%	98.45%
`+m1p=1`	+ ownership scaled by p	48.07%	45.53–50.60%	99.4	97.33%	98.72%
`+norepeat=1`	+ repetition guard	43.53%	41.13–45.93%	99.5	98.45%	98.63%

no honest opponent, against the 4.4 points §5.4 records on the mixed panel.

The parameter is not identified, and was not tuned. A sweep of `priorTheta` over 5 values, $\{0.20, 0.26, 0.35, 0.445, 0.60\}$, at 2 seeds does not identify it: the ordering of the settings is not stable between the two banks and every pairwise difference sits inside the cluster-bootstrap interval. The fitted value was therefore retained rather than adjusted, because adjusting it on this evidence would be selection on noise. The principled repair is M7 — a per-seat online type posterior with data-biased shrinkage, so that a certificate from a seat whose type posterior has drifted toward "deceptive" is discounted — and M7 is specified and not built (§5.2).

Consequence for the worst case. Across the full 12-style set — the 9 of Table 8 plus these three — the lowest cell FishBot v0.5 records anywhere is 50.96% against the feint, on the weaker of the two deception banks, marginally below its mirror cell of 51.61%. Which of the two is *the* worst case is not resolved by these banks: averaged over both deception banks the feint reads 51.52%, which is above the mirror rather than below it, and the two candidates sit within half a point of each other either way. The honest statement is that FishBot v0.5's floor over the twelve styles is about 51.61%, attained on the mirror or on the feint depending on the bank. FishBot v0.4's worst case remains its own mirror at 50.00%. Neither policy falls below a coin flip against anything in the set, and no aggregate over it should be quoted without this table and Table 8 beside it.

7.7 Mechanism ablations

Every row of Table 10 is a frozen-parameter ablation: the fitted vector is FishBot v0.5's in all of them, and no row was refitted around the mechanism it removes. A cheap removal may still be doing work a refitted policy would find another way to do, and an expensive removal may be expensive only because the surrounding weights were fitted to rely on it. That qualification applies to every row and is not repeated per row; a matched-budget refit is one of the experiments §9 names as not run.

Stage 2 is dead code once M1 is on. The `m1=1` and `m1=1,stage2=0` rows are identical to the digit in win rate, events, declaration accuracy and their opponent's declaration accuracy, and so are

Table 11: Aggregate forecast reliability for FishBot v0.5 (E6). Population: the calibration bank of 400 deals, seed 717171, FishBot v0.5 against FishBot v0.4, weights frozen; the unit is one forecast and n counts forecasts of that type. Voluntary and forced declarations are pooled. All entries are point estimates: no clustered uncertainty is computed for Brier score, log loss or expected calibration error, so small differences between them are not resolved by these numbers. Source: `research/v05/results/E6-calibration-v05.txt`.

Forecast	n	**Brier**	**log loss**	**ECE**	**mean pred.**	**mean obs.**
ask succeeds	36,502	0.12219	0.36584	0.02231	53.2%	53.3%
declaration correct	3,620	0.01646	0.06247	0.01493	96.7%	98.2%

the `m1=1,m2=1` and shipped rows. This is not a coincidence of the bootstrap: with the live-ask gate on, games end at about 99.7 events and the event-count escalation's second stage is never reached, so switching it off changes nothing that happens. The measured cost of 8.13 points that §4.6 attributes to that rule is a cost it can only levy on a policy that deadlocks.

The rows with the best win rates are the rows that still deadlock. M8 alone is the largest number in the table, 56.60%, and M2 + M8 the second largest, 56.33%; both play games of about 142.3 events against the control's 141.8, so both retain the failure mode in full. They win by declining to cash a declaration the policy itself scores near zero: their own declaration accuracy rises from the control's 89.10% to 96.67%, while FishBot v0.4's in the same games falls from 88.61% to 83.55% — an asymmetry this experiment records without explaining, since the two arms play games of nearly the same length. This is the forcing dilemma of §4.6 measured again inside FishBot v0.5's switch set: patience without termination buys points and keeps the pathology, and the configurations that terminate (99.7 events, 98.47% own declaration accuracy, 98.45% for the opponent) are the ones that give the points back. The mechanism package is bought with win rate, not for it.

Two designed mechanisms were rejected on measurement. Both are retained as switches, both ship off, and both are recorded here so they are not rebuilt. Scaling the four ownership features of §5.1 by hit probability on top of the gate — `m1p`, which is a switch and not part of the shipped configuration — costs −1.33 points relative to the shipped configuration, and was measured at −5.6 points on top of M1, M2 and M8 at design time, which is why the switch ships off. Its own diagnostic points at the reason: it *raises* the hit rate, from 53.60% to 54.63%, while losing games. Once the gate has removed the $p = 0$ case there is no dead ask left for the scaling to suppress, so all it does is rescale live candidates against one another, and it buys immediate hits at the expense of the ownership progress the rest of the linear score was fitted to trade against.

The repetition guard — `norepeat`, forbidding a seat from asking the same (card, target) pair twice — costs −5.87 points, and was measured at −6.0 points on top of M1, M2 and M8 at design time. It also ships off, and it has an explanation worth stating, because the guard looks like a structural backstop that could not hurt. The reason it does is that cards move: forbidding a (card, target) repeat forbids asking for a card the actor has just watched transfer to that opponent in a public event, which is frequently the best ask on the board. The cost shows in the hit rate, which falls from 53.60% to 52.85%, and not in the event count, which is unchanged at 99.5 against 99.7: the guard is not preventing a deadlock — there is none left to prevent — it is deleting asks the policy would otherwise land. And the case it was designed to catch is already covered: a repeat of an ask that missed is provably dead in the sense of Definition 1, so M1 removes it without a separate rule, which is why the guard measured as exactly neutral when tested alone in Table 4 and negative once M1 was present.

Table 12: FishBot v0.5 against FishBot v0.4 under four rules dialects (E7). Population: 250 deals in six seat rotations, 1,500 games per row, seed 828282, both policies frozen. "Forced rate" is FishBot v0.5's forced declarations per game and the two accuracy columns are the share of each team's forced declarations that were correct. The FishBot v0.3 legacy dialect implies declaration on turn only, so its rows track the no-out-of-turn dialect. In the default and no-cardless dialects a forced declaration occurs a handful of times per row, so those two accuracy cells rest on single-digit denominators and should not be read as rates. Intervals are deal-clustered percentile bootstrap. Source: `research/v05/results/E7-rules.jsonl`.

			forced	forced correct	
Dialect	**FishBot v0.5 win**	**95% CI**	**per game**	**FishBot v0.5**	**FishBot v0.4**
default (§2)	50.20%	47.87–52.53%	0.0040	33.33%	0.00%
no out-of-turn declaration	51.53%	49.00–54.00%	0.1973	97.97%	88.85%
no cardless declaration	50.47%	48.13–52.80%	0.0040	33.33%	0.00%
FishBot v0.3 legacy	51.33%	48.80–53.80%	0.1973	93.58%	88.89%

7.8 Calibration

Neither aggregate in Table 11 characterises the estimator it scores, for the reason the v0.4 study gave: both are dominated by one bin. Of 3,620 declaration forecasts, 86.4% fall in the top decile at 99.9% predicted against 99.9% realised, which is most of what the expected calibration error of 0.01493 measures. The decision-relevant bins run the other way: in $[0.7, 0.8)$ the declaration forecast is *under*confident, 75.7% predicted against 89.1% realised on 192 forecasts: the estimate the stopping rule thresholds is conservative in exactly the range where the threshold binds.

The ask forecast is the one this report has a reason to look at, because M1 gates on a hard deduction while the ask score prices everything above it. Its aggregate is close to unbiased — 53.2% mean predicted against 53.3% realised over 36,502 forecasts — and it is overconfident precisely at the bottom, where the lowest decile predicts 4.6% against 1.8% realised on 3,529 forecasts, and underconfident in the middle, where $[0.4, 0.5)$ predicts 45.0% against 51.8% on 2,805. The gate itself is unaffected, since it is a deduction and not a forecast; what the bottom-decile bias means is that among the asks that survive the gate, the near-dead ones are still priced somewhat too high, which is a reason to expect the M3 net-information term to have something to price rather than a defect this report repairs.

7.9 Rule dialects

The head-to-head margin is insensitive to the dialect: the four rows of Table 12 span 50.20% to 51.53%, a range narrower than the width of any one of their intervals, and every interval overlaps every other. None of them separates from 50%; the closest is the no-out-of-turn row, whose interval reaches down to exactly 50%. The v0.4 study's dialect experiment moved by several points, but it measured a different pairing on a different bank, so the two are not comparable; all that is claimed here is that the FishBot v0.5-against- FishBot v0.4 margin does not depend on which of these four dialects is played.

What the dialect does change is the population of states the mechanisms act on. Disabling out-of-turn declarations raises the forced-endgame rate by a factor of about 49, from 0.0040 declarations per game to 0.1973, because a team that may only declare on its own turn arrives at the cardless endgame far more often. On that population FishBot v0.5's forced declarations are correct 97.97% of the time against FishBot v0.4's 88.85%, and FishBot v0.4's overall declaration accuracy falls to 95.46% against FishBot v0.5's 98.63%. The absolute accuracies are much higher than the 25.00% of Table 6 because the states are not the same states — a forced endgame reached because declaration was illegal until now is far more determined than one reached after a full game of asking — so the two measurements are not comparable in level. Nor are they comparable in the size of the gap between the two policies: the gap here, 97.97% against 88.85%, is less than half the one Table 6 records between 25.00% and 0.14%. What transfers across them is only the sign of that gap, which is what M2 predicts, and that sign is the only claim made from this row. Ask hit rate also falls sharply under these dialects, to 44.97% from 53.77%, which is a property of the dialect and not of either policy.

7.10 Throughput

FishBot v0.5 plays 282 self-play games per second over a timing run of 600 games, a wall-clock rate for one run with no interval. The more useful comparison is the matched one the ablation bank supplies, because it holds the opponent, the deals and the machine fixed: on that bank the mechanism-disabled control of Table 10 runs at 160 games per second and the shipped configuration at 280, a factor of 1.8×. (The two figures differ from the self-play number above because they are played against FishBot v0.4 rather than a mirror.) The failure mode of §4.2 is therefore a compute cost as well as a play defect: the deadlock is played out move by move and every event in it runs a belief update. The factor exceeds the event ratio — 141.8 events per game against 99.7 — which is consistent with the certificate accumulation §9 notes, but this timing is not instrumented finely enough to isolate that. Every evaluation in this section completes in minutes at these rates; the fit of §6 does not, and its cost is not reported here.

8 Corrections to the v0.4 study

The v0.4 study [2] and its supporting artifacts contain claims that the measurements of §4 contradict. They are stated here as corrections, in one place, rather than quietly patched. Each entry gives what was claimed, where, what is now measured, and what replaces it. Where the v0.4 *paper* and a v0.4 *artifact* disagree with each other, that is said explicitly, because in the load-bearing case the paper is right and the artifact is wrong.

C1. The information-freeze account of non-termination is withdrawn. The v0.4 termination artifact, `research/v04/results/E11-termination.md`, states that the locked-half-suit theorem implies that such a half-suit is frozen "and, for the same reason, that no further information about its allocation can ever arrive," and concludes that two policies which both decline to cash an uncertain half-suit therefore deadlock. It repeats the claim as "Theorem 1 freezes information as well as ownership," and closes by offering that proposition as a contribution to the rules literature. That is false, and all three statements are retracted.

It is false at the level of the rules before any measurement: an ask inside a half-suit locked to the asker's own team satisfies every legality condition, and the deduction machinery publishes the asker's own exclusion whether the ask succeeds or not. It is false as measured: 68.5% of such asks strictly raise a teammate's exact probability of the correct allocation, mean +0.096, maximum +0.664, and 59.9% strictly lower the exact marginal entropy, by a mean of 0.325 nats (§4.4). And it is not the mechanism of the observed deadlock in any case: in 66% of long games no half-suit is locked to any team at any point in the dead run, and only 3.2% of long games have their entire cycle inside a lock. What replaces it is §4.3: the deadlock is a deterministic two-question cycle at a fixed point of the public information state, produced by an ask score in which ungated ownership features outbid the hit-probability term at $p = 0$.

The v0.4 *paper* is not implicated in this error and should not be read as if it were. Its Observation on locked half-suits states the correct position — that ownership freezes while allocation information need not, that asks inside a locked half-suit stay legal and carry certificates, and that ownership monotonicity alone establishes neither informational freezing nor non-termination. The error is confined to the artifact, was flagged in the engine's own specification document, and was never corrected there.

C2. The reported termination property was purchased, and the price was not reported. The v0.4 study reports an action-limit rate of 0.000% across its experiments and describes the graduated event-count escalation as the mechanism. Both statements are accurate. What was not reported is what the escalation does in the population where it actually fires. In mirror play 384 declarations are made at or after the forcing horizon and 58.59% of them are wrong, against a baseline declaration error of 10.44%; bucketed by the confidence the policy itself attaches, 130 declarations in 200 games are made below a self-assessed probability of 0.1 and every one is wrong. The escalation's second stage returns "declare" before inspecting the allocation probability at all, and removing it is worth 8.13 points against a mirror opponent while changing nothing against any weaker style tested. A zero cap-hit rate against a panel of weaker policies is therefore not evidence that termination was solved; it is evidence that the horizon was rarely reached in that panel. The correct statement is that FishBot v0.4 converts non-termination into misdeclaration, and that the conversion rate is a strong-opponent tax.

C3. Forced-endgame declarations were reported as inaccurate and recorded as a known gap; they are a certainty of loss with a mechanical cause. The v0.4 evaluation tables carry a forced-declaration accuracy of zero, and the engine specification records that the allocation routine takes a per-card argmax with no capacity constraint. The two were never connected. §4.7 connects them: over 6,000 distinct mirror games, all 281 forced declarations violate the declarer's own capacity constraint, all have exact posterior probability zero, and the declarer's own stated confidence is exactly zero in every one. The best feasible allocation would have been correct in 40.6% of them. This is not a hard inference problem being lost; it is a policy naming an allocation it has itself scored at zero. Two notes attach to the same measurement, neither of which touches the rate. The count of 281 corrects, downward by half, the figure the diagnosis run first reported: in a mirror match the six seat rotations are three deal shifts times two team orientations, both orientations play byte-identical games, and every such run splits its forced declarations exactly evenly between them. And the result does not rest on that run alone — an instrument sharing no code with it, recomputing each declaration from the opening deal, reproduces it at five further mirror seeds.

C4. The known gap about the value-function coefficients has the sign backwards. The v0.4 specification records that the sixteen compiled value coefficients are not the ones in the fitting artifact, framed as a defect. The compiled vector is the better of the two: substituting the fitted vector costs 1–5 win-rate points, including 2.67 points against FishBot v0.3, with the sign stable across nine opponents and three independent seed banks. Two riders on that figure are corrections to the present study rather than to the v0.4 one. The size is not uniform across the panel: at a fresh seed the mirror is the *smallest* effect and the only one whose cluster-bootstrap interval includes zero, so a narrower per-opponent range stated in an earlier draft of this analysis, and the reading that the compiled vector's advantage is largest in mirror play, are both withdrawn. The real defect is larger and different. The model is effectively one feature: held-out R^2 is 0.2327 for all sixteen and 0.2354 for a bias term plus the score differential alone, two features are algebraically identical, and five more cancel at every call site — moving all five to ± 9 changes none of 31,788 declaration decisions. The v0.4 limitation that no goodness-of-fit figure was reported for the deployed coefficients is upgraded here to a measurement, and it is not flattering.

C5. The policy prior is one channel, not two. The v0.4 study describes the prior as two coefficients, a weight on having asked in a half-suit and a weight on having taken turns without asking there, and reports a frozen-policy ablation for the pair. The second is not an independent channel: the exponent rearranges so that its term is card-independent, and the capacity normalisation then removes it. Deleting it costs nothing measurable — 49.00% [45.8, 52.2] against the honest mirror, a paired delta of +0.3 points across a four-opponent panel, and a fifteen-fold sweep flat inside its intervals. The ablation result for the pair stands, and the attribution of it to two channels does not. This also corrects a hypothesis of the present study rather than of the v0.4 one: the silence channel that the project owner's manoeuvre attacks was not misreading silence, it was not reading it at all.

C6. Asks inside a locked half-suit: both halves of the v0.4 hedge are now measured, and they point in opposite directions. The v0.4 limitations describe these asks as "a channel the fitted configuration has no term for . . . unexploited here, not shown to be worthless." They are not worthless: §4.4 prices them, and a greedy ladder of them reaches certainty in 82.9% of attempts at an information exchange rate of about 37 : 1 in the sender's favour. They are also not usable in the form the hedge implies. Selecting them requires the actor to know its team owns the half-suit, and that condition is provable from public information at only 0.34% of mirror decisions; at the deadlock states the exact probability an owning team assigns to its own ownership averages 0.068 and peaks at 0.342, so not one of the 36 pairs measured comes near the 0.999 bar. The channel has to be priced against ownership probability rather than gated on ownership. The first pass of the present study wrote "provably owns" where it meant "in fact owns," and that wording is corrected here.

C7. The eight-level forced-endgame willingness ladder does not function, and leaks more than the rules license. The v0.4 dialect table presents the ladder as a graded willingness mechanism and contrasts it with v0.3's two-level version. It is inert: the statistic it thresholds is exactly zero in 210 of 210 surveyed (player, half-suit) pairs, a consequence of C3, and four ladder shapes including one with no rungs at all produce bit-identical output. Separately, a ladder of this shape transmits 2.19 bits per candidate, against the roughly one bit of willingness the rules make public. M2 restores the ladder's function for free; the leak figure is the reason §5.4 declines to extend the same device to mid-game turn transfer, where the audience still holds cards.

C8. Waiting was free by construction, so observing that waiting dominates proves nothing. The v0.4 declaration rule compares the value of acting against the value of waiting, and the waiting branch is evaluated by passing all-zero deltas to a value model in which the public event count appears nowhere. The value of waiting at time t therefore equals the value of waiting at $t + 1$ as an identity of the implementation, not as an approximation or a finding. Any argument that patience weakly dominates in that model is self-fulfilling, and the only thing that ever broke the tie was the hand-set horizon whose cost C2 records. §5.4 reports that the literature's remedy for this — a strictly positive holding cost — was implemented and swept, and that it shifts the mean without touching the tail, so the correction here is not that v0.4 chose the wrong cost but that the well-posedness of its stopping problem came from a move cap rather than from anything in the model.

C9. Scope: the published head-to-head result stands; its generalisation caveat now has an instance. FishBot v0.4's reported 75.07% against FishBot v0.3 is not contradicted by anything here, and the v0.4 study was explicit that it established nothing about generalisation to unseen opponent populations. §4.2 supplies a concrete instance of that caveat biting: on the same instrument, against a population the panel did not contain, the dead-ask rate differs by a factor of fourteen and a third of games contain an extended run of guaranteed misses. The lesson is about evaluation design and is acted on in M10 — the mirror belongs in the fitting panel and the pathology statistics belong in the commit gate.

9 Threats to validity

This section is in two parts. §9.1 names the eight limitations that bound what §7 may be read to show, each with the experiment that would resolve it and none of which has been run. §9.2 groups the remaining threats by the kind of validity each bears on. Caveats stated in full at the point of measurement are cross-referenced rather than restated.

9.1 Named limitations, and the experiment each needs

L1. The strength gain is small, and often within noise. Head to head FishBot v0.5 scores 51.61% against FishBot v0.4 [49.28–53.94] — +1.61 points on FishBot v0.4's 50.00% mirror baseline, on an interval that contains even money. The nine-style means are level at 83.65% against 83.57%, and on minimax regret over that style set FishBot v0.4 is marginally the better of the two, 2.06 against 1.67. Most of the per-style differences are a point or less — hunter 0.00, bluffer −0.06, random 0.00, detective −1.67, diversifier +1.22, FishBot v0.3 −1.06 — and the largest, FishBot v0.2 −1.33 and lockout +2.06, are of a size the per-cell intervals do not separate. No style in the table is one on which the two policies' intervals fail to overlap. Nothing in this report should be read as FishBot v0.5 playing better than FishBot v0.4; the case for it is the pathology table and the deception panel. *Resolving experiment:* a paired common-deal evaluation at an order of magnitude more deals per cell, with a per-style optimum computed for each style, so that regret is measured against a best response and not against the better of two arms [28].

L2. No exploitability probe was run against FishBot v0.5. The v0.4 study ran one; this one did not, and the battery of §7 contains no best-response search of any kind. FishBot v0.5 therefore inherits no robustness property from FishBot v0.4's probe and claims none of its own. This is a strict scope limit rather than a hedge: the deception panel measures performance against three *fixed* deceptive rules, which is not the same quantity as performance against a policy that is searching for this policy's worst case. *Resolving experiment:* a local best-response probe against FishBot v0.5, reported as an achieved score that is a lower bound within its search class [45], run with and without M6, since a deterministic argmax sits at the deterministic leakage bound whatever its feature weights [15].

L3. Mechanisms M3–M7 are specified and unbuilt, and two patches for them are unmeasured. §5.2 specifies the net-information ask term, the per-seat knowledge model, target-dimension selection, partner-aware stochastic selection and the online opponent model. None is on FishBot v0.5's decision path. Patches for M4/M5 and for M7 exist in the repository under `research/v05/patches/`; neither has been evaluated in play, and no number anywhere in this report is attributable to either. They are recorded because unmeasured code that looks finished is a hazard, not because they support a claim. *Resolving experiment:* apply each patch, measure it alone against the frozen configuration and then jointly, in the build order of §5, refitting after each — since §5.1 already shows frozen-parameter ablations understating a mechanism whose parameters were fitted for a different policy.

L4. Forced-endgame accuracy is still far below the feasible ceiling. M2 takes forced-endgame accuracy from 0.14% to 25.00% over 24,000 games per arm, and the measured ceiling for the best capacity-feasible allocation at those states is 40.6%. The gap is not closed, and the reason is that feasibility is exact while the score that ranks the survivors is not: it runs on the deployed approximate belief. *Resolving experiment:* a calibration study of M2's allocation estimate against exhaustive enumeration on small reachable states, to bound the gap between its exact feasibility test and its approximate score, followed by re-ranking survivors with the exact belief where the state is small enough to afford it.

L5. A single fitting round. The configuration evaluated here comes from 1 round of 40 generations (§6); the v0.4 configuration came from 5 rounds. The gap between in-panel and held-out performance is already visible in the one round that was run: the best generation scored 57.00% against FishBot v0.4 inside the panel, while the frozen configuration — the distribution mean the winner's-curse guard preferred to it (§6.4) — scores 51.61% against FishBot v0.4 on a bank the fit never saw. The held-out figure is the one reported, and the difference is a regression to be expected of a single round rather than an anomaly. *Resolving experiment:* further rounds on fresh deal banks, each evaluated on a bank the fit never saw, with the round-by-round in-panel and held-out figures published together so the regression can be read directly.

L6. The feint exposure is real, replicated, and not addressable by retuning. Against the feint archetype FishBot v0.5 scores -2.2 points relative to FishBot v0.4, and does so at both seed banks (§7). The proximate cause is identified: the fit raised the weight on "this player asked here" from 0.264 to 0.445, and §4.8 establishes that the exposure is over-weighting the policy prior rather than weighting it at all: against the deceptive archetypes themselves, deleting the prior outright is worse still, by 4.60 points [2.63, 6.58]. A sweep of that weight over 5 values $\{0.20, 0.26, 0.35, 0.445, 0.60\}$ at two seeds does not identify it: the ordering is unstable across seeds and every pairwise difference sits inside the cluster-bootstrap interval, so the fitted value is retained rather than tuned on noise. *Resolving experiment:* M7, a per-seat online type posterior with data-biased shrinkage [22], evaluated against the deception panel at the seeds used here plus fresh ones, and reported as a frontier rather than an operating point [21].

L7. Owner decision D2, partner-aware play, is specified and not implemented. The design requires two regimes — convention-rich, lower-temperature play with bot teammates, and grounded, less predictable play with human or unknown teammates — reported separately, with the self-play configuration never headlined as the one a human will meet (§5.2). FishBot v0.5 implements neither. It is a single deterministic policy used in both regimes, and every number in this report is from the self-play regime. There is no human-teammate evaluation of FishBot v0.5 at all, which matters most for the one result this study did not obtain by measurement: the report that a well-trained human outperforms FishBot v0.4 on some reasoning. *Resolving experiment:* build M6 with the temperature and the convention flag keyed on the partner regime, then evaluate the two regimes as separate arms — bot-teammate self-play, and FishBot v0.5 partnered with a policy it was not trained with as a stand-in for an unknown teammate — before any human-teammate sessions are attempted.

L8. Owner decision D1, conventions behind a flag, has no flag. §5.3 states the policy: build the signalling machinery behind an explicit switch, ship it off, and publish the with/without difference as a result in its own right. The switch does not exist. No codebook, no convention machinery and no receiver-side decoder were built, so the difference that was to be published is unmeasured, and the headline configuration is "conventions off" only in the trivial sense that there is nothing to turn on. *Resolving experiment:* implement M5's codebook term behind `-conventions`, and report the paired difference between the two settings on the same deal bank, separately for each partner regime of L7, since a codebook is worth nothing to a teammate who does not share it.

9.2 Threats grouped by kind of validity

Internal validity. Every ablation in Table 4 is a frozen-parameter ablation: no row was refitted, so no row measures a mechanism's standalone value under parameters chosen for it. The causal isolation of §4.5 was performed with a prototype filter applied to the FishBot v0.4 policy, not with the shipped M1, and the two differ in the fallback rule and in the ownership scaling. The diagnosis probes are copies of the shipped policy rather than the shipped policy itself; equality with the original was checked by reproducing its aggregate statistics to the reported digits after every instrumentation change, which is a strong check and not a proof of identity. On the other side of the ledger, the evaluated configuration is the frozen one: the round-trip assertion that the compiled defaults and the fitted vector name the same policy passes, and the rules audit reports 0 violations in 23,594,580 checks with determinism reproduced.

Construct validity. The provably-dead predicate of Definition 1 is a hard deduction and admits no false positives, but several of the headline statistics that motivate M3 are ground-truth statistics: the lock census, the "asks inside a half-suit the team already owns" rate and the certificate-ladder measurements all classify half-suits by the true deal, not by what any observer can establish. §4.4 and §8 state the consequence — the provable version of the condition holds at 0.34% of decisions — but the reader should not read those rates as an achievable policy. M2's allocation probability is computed with the deployed approximate belief, so it is an estimate; the feasibility constraints it enforces are exact, the score it ranks survivors by is not. More generally, the posterior throughout this work is exact with respect to the rules and a uniform prior over consistent deals, which is the policy-agnostic object; the true posterior weights deals by opponent policy, and the two-scalar prior examined in §4.8 is a parametric stand-in for that correction rather than the correction — which is also why L6's exposure is a modelling gap and not a tuning error. Finally, win rate does not capture declaration reliability, calibration, or the pathology statistics that gate this work, which is why they are reported separately.

External validity. Every opponent in this study was written inside this project or ported from an earlier FishBot. The deceptive archetypes of §4.8 are commitment deceivers — they follow a fixed deceptive rule for the whole game — and while the withholder is a direct model of the manoeuvre the project owner reported, it does not reproduce what he actually did, which was to adapt within a game and across games. The gain of +7.2 points against it is therefore a result about a fixed rule, not about a human. The report that a well-trained human outperforms FishBot v0.4 on some reasoning rests on one player's account of live sessions, is not a controlled measurement, and is treated here as a hypothesis that generated experiments rather than as a result. No expert human logs exist for this dialect and no independently authored engine for it was located, so nothing here tests the agent against a strategy source outside this project. One point runs the other way and is worth stating: the three deceptive archetypes, together with FishBot v0.2, the bluffer and the random opponent, were absent from the fitting panel (§6), so the deception panel is a genuinely held-out measurement of deal bank *and* opponent style, which most of the nine-style table is not.

Statistical validity. The per-style table is a single seed bank (300 deals in six rotations per cell, seed 515253), and the deception panel is two; only the FishBot v0.4 pairing has been run at more (5 banks, 1,500 deals, mean 50.77%, range 50.06–51.39%). Several of the diagnosis figures are stable in rate and unstable in magnitude across seeds, and the pooled values are used above for that reason. Rerunning the information measurement at two fresh seeds reproduced the rate (68.5% pooled against 70.6% at the original seed) but moved the mean improvement from +0.126 to +0.096 pooled; the certificate ladder reached certainty in 21 of 21 attempts at the original seed and in 82.9% pooled, at a median of 4 asks rather than two; and the ratio at which a teammate's best allocation moves toward the truth rather than away from it fell from 14.3 : 1 to 3.3 : 1 pooled. All keep their sign. The "information-free at every state" result of §4.5 is scoped to 42 states already inside a repeat cycle and is not a claim about FishBot v0.4's asks in general — indeed 16.46% of all its asks land inside a half-suit its team already owns and do emit certificates, by accident rather than by choice. Intervals quoted are deal-clustered percentile bootstrap intervals, which under-cover in evaluations of this kind [46].

Opponent-population dependence. Nothing in this report bounds exploitability, and no such bound is claimed; L2 is the scope statement and [45] the form the eventual measurement must take. Two of the mechanisms that would bear on it are unbuilt: while action selection remains a deterministic argmax, leakage sits at the deterministic bound whatever the feature weights [15], so any leak penalty in the ask score is acting on the wrong variable; and the opponent model remains a fixed global prior rather than a per-seat one, so the robustness frontier that M7 is supposed to expose does not yet exist to be reported.

Rules-dialect dependence. All results are for the dialect of §2, which is one explicit selection among documented variants [3]. Arbitration by lowest seat is a modelling choice rather than a rule, and the action cap is a backstop rather than part of the game. The deadlock measurements in particular are sensitive to the cap only in the patient configuration, where the cap is what ends the game.

Computational and model-class limits. The ask score remains a twenty-feature linear function refined by a one-ply expectimax, which caps what any reweighting can express; M3 and M5 add terms to that class rather than replacing it. The value model inside it is effectively one feature — held-out R^2 of 0.2327 for all sixteen against 0.2354 for a bias term plus the score differential alone, where a tree reaches 0.388 — so M9's decision to rebuild or retire it is still open, and until it is taken the declaration rule is scoring on a model whose capacity has been measured and found wanting. M2's search is exhaustive over at most 3^6 assignments and returns no candidate at all when an opponent provably holds a card of the half-suit or when no teammate may hold one, so the declaration path still needs a defined behaviour at those states. The duplicate-certificate growth noted in §4.3 makes the exact belief progressively more expensive inside a long cycle without changing what it computes; de-duplication is an unbuilt engine improvement.

Experiments not tied to a single limitation above, none of which has been run: a provability oracle over the transportation polytope, to establish how far the 0.34% provable-ownership rate can be widened, which decides whether M3 can ever gate rather than price; and an adaptive human-like opponent, or logged human play, to test whether the owner's live result survives a controlled setting that commitment deceivers do not reproduce.

Provenance of the figures in this report. Of the 574 distinct quantities quoted in the text, 409 are emitted directly from an experiment artifact by `engine/build_tables_v05.py` and cannot drift from it; the remaining 165 are transcribed out of the diagnosis reports of §4 rather than regenerated. Transcription is the weaker guarantee, and it is the mechanism by which a number goes stale when an artifact is regenerated and the prose is not. It is constrained rather than eliminated: `paper/check_provenance.py` requires every transcribed quantity to sit under a comment naming the artifact it came from and fails the build if that artifact is absent, and it reports the split above so that this paragraph cannot itself go stale. Extending the generator to parse the diagnosis reports would close the gap; it has not been done, and until it is, the 165 transcribed figures carry the same standard of evidence as the v0.4 study's hand-entered numbers, which is precisely the standard §8 criticises elsewhere.

10 Conclusion

FishBot v0.4 was published on the strength of a panel of weaker opponents. Played against a copy of itself it spends 39.04% of its asks on cards it can *prove* the target does not hold, repeats 40.03% of its asks exactly, and gets 10.44% of its declarations wrong. This report measures that failure, corrects the account the v0.4 study gave of it, and reports the policy that removes it. What follows is what the evidence in §4 and §7 supports, stated without extension.

The deadlock is a fixed point of the ask policy, not a freeze of information. In a game where nothing compels a claim, the natural explanation for two patient policies stalling is that they have run out of things to learn. That explanation is wrong here twice over. It is wrong about the mechanism: the observed stall is a deterministic two-question cycle, present in 12 of 14 long games at the forensic seed, and across the pooled seed banks no half-suit is locked to any team at any point in the cycle in 66% of long games. And it is wrong about information: where a half-suit *is* locked, 68.5% of the asks its owning team may legally make inside it strictly raise a teammate's exact probability of the correct allocation, by a mean of 0.096 and up to 0.664. Ownership monotonicity does not imply informational monotonicity, and the two were conflated.

The practical consequence is that the repair is a subtraction, not a forcing rule. Declining asks whose success probability is exactly zero — a hard deduction from the actor's own public knowledge, which cannot introduce a modelling error — takes the dead-ask rate from 39.04% to 0.01%, the share of games containing a run of six consecutive dead asks from 34.33% to 0%, and the longest such run from 286 to 1. It does so with the event-count escalation that used to force claims switched *off*: declarations made at or after the old horizon fall from 384 to 0, declaration error from 10.44% to 2.07%, and no game in either arm reaches the action cap. Forcing claims was never the fix; it was the mechanism that converted the stall into misdeclaration, at a measured cost of 8.13 points against a mirror opponent.

A per-card argmax allocation is infeasible, not merely unlikely. The second defect is of a different kind from a low-probability guess, and the difference is worth stating exactly. FishBot v0.4 named its forced-endgame allocation by taking, card by card, the teammate with the highest marginal — a procedure with no capacity constraint, which can therefore give a teammate more cards than that teammate holds. All 281 such declarations over 6,000 distinct mirror games were wrong, and the reason is not that the estimate was poor. The named allocation has posterior probability exactly zero: it is excluded by the declarer's own arithmetic, not merely disfavoured by it. A policy losing a hard inference problem and a policy naming an outcome it has itself ruled out are different failures and need different repairs. Searching the capacity-feasible set instead takes forced-endgame accuracy from 0.14% to 25.00% over 24,000 games per arm — against a measured ceiling of 40.6% for the best feasible allocation at those states, so the gap is narrowed and not closed. The state is also reached 6.3× less often, but that belongs to M1 rather than to M2: what was stripping a team of cards while half-suits stood undeclared was the runs of guaranteed misses, not the declaration rule (§7.3).

What the v0.4 study said that is withdrawn. §8 is the register. The load-bearing entry is C1, the information-freeze account of non-termination, which is retracted with the measurement that refutes it; the v0.4 *paper* states the correct position and is not implicated, while the supporting artifact states the wrong one. The remaining entries correct the reported termination property (purchased, and the price unreported), the status of the forced-endgame accuracy figure (a mechanical certainty, not a hard problem lost), the sign of the value-function "known gap," the claim that the policy prior is two channels, the usability of asks inside a locked half-suit, the willingness ladder, the well-posedness of the stopping problem, and the scope of the published worst case. Three of those entries also carry corrections to earlier drafts of the present study rather than to the v0.4 one, and are recorded in the same place for the same reason.

What FishBot v0.5 costs, and what it does not buy. It does not buy win rate. Over the nine-style panel of §7 (300 deals in six rotations per cell, seed 515253) the means are level — 83.65% for FishBot v0.5 against 83.57% for FishBot v0.4 — and head to head FishBot v0.5 scores 51.61% [49.28–53.94], which is +1.61 points on FishBot v0.4's 50.00% mirror baseline on an interval that contains even money. Across 5 further seed banks the same pairing averages 50.77% with a range of 50.06–51.39%. On minimax regret over the style set FishBot v0.4 is marginally the better of the two, 2.06 against FishBot v0.5's 1.67. Per style the differences run in both directions: +2.06 on the lockout opponent, +1.61 on the mirror and +1.22 on the diversifier against −1.33 on FishBot v0.2 and −1.06 on FishBot v0.3, with the remainder at half a point or less and no per-cell interval separating either policy from the other on any style. FishBot v0.5's worst case over the nine styles is 51.61%, against FishBot v0.4, and FishBot v0.4's is 50.00%, against a copy of itself. In both cases the worst cell is the opponent of the policy's own strength, which is the worst case for any policy in this family, and which is why the v0.4 study's floor of 75.07% describes a panel rather than a policy. The case for FishBot v0.5 is the pathology table and the deception panel below. It is not the win rate, and this report does not offer one.

The result that speaks to the incident that started this. The work was prompted by a live report: the project owner held cards of a half-suit he had been asked for and then declined to ask back in it, and FishBot v0.4 was visibly confused. Modelled as an archetype — after being asked in half-suit S while holding other cards of S, avoid asking in S for the next several turns — FishBot v0.4 wins 65.35% of its games against that archetype and FishBot v0.5 wins 72.53%, a gain of +7.2 points replicated at 2 independent seed banks of 2,400 games each. The silent archetype moves the same way, by +2.3 points.

The mechanism is worth being precise about, because it is not the one the design document proposed. FishBot v0.5 has no opponent model that FishBot v0.4 lacks: M3–M7 are specified and unbuilt, the policy prior is still fixed and global rather than per-seat, and nothing in FishBot v0.5 watches a seat and updates a type. What changed is that FishBot v0.5 never spends a turn on an ask it can prove will miss. A misleading *absence* of asks works by making a policy's own inferences point at the wrong half-suit; a policy whose candidate set is already pruned to asks with strictly positive hard-consistent probability gives that misdirection much less to move. Robustness here was bought by removing a defect, not by adding a model — which is the more encouraging of the two possibilities, and also the one with a lower ceiling.

The same panel records a loss. Against the feint — which manufactures a *false* ask-legality certificate rather than withholding a true one — FishBot v0.5 scores 51.52% against FishBot v0.4's 53.71%, a difference of −2.2 points that replicates at both seeds. It is not an accident of tuning noise: the fit raised the weight on "this player asked here" from 0.264 to 0.445, and the diagnosis had already established that over-weighting the policy prior is exactly this exposure — while deleting the prior outright is worse still against the deceptive archetypes themselves, by 4.60 points [2.63, 6.58]. Across the full twelve-style set FishBot v0.5's worst case is the feint at 50.96%, marginally below its mirror. No aggregate figure in this report should be quoted without that table beside it.

What comes next, in order. The ranking follows from the exposure above rather than from the design document's build order.

1. **M7, the online per-seat type posterior.** This is the principled fix for the feint, and §9 shows why nothing cheaper will do: a sweep of the prior weight over 5 values at two seeds does not identify the parameter. Shaped as a data-biased response [22], a manufactured certificate from a seat whose type posterior has drifted toward deceptive is discounted rather than believed, and the exploitation/robustness dial becomes explicit and reportable [21] instead of being frozen at a fitted

constant.

2. **M4, M5 and M3, in that order**, because each is the precondition of the next. A per-seat knowledge model is nearly free from a public transcript; it makes the target dimension scoreable, which is currently discarded outright — the ask score opens by voiding its target argument, throwing away a mean of 0.639 free bits per ask, with at least two hard-indistinguishable targets on offer at 46.6% of decisions and roughly 41 bits per team per game in total; and only then can the net-information term of M3 be priced rather than gated, which §4.4 shows is the only form in which it can ever fire.
3. **An exploitability probe.** The v0.4 study ran one; this one did not, and until it does FishBot v0.5 claims no robustness property that a best response could contradict. Such a probe reports an achieved score that is a lower bound within its search class [45], and it is worth running both with and without M6, since a deterministic argmax sits at the deterministic leakage bound whatever its weights [15].
4. **A value model that is more than one feature.** Held-out R^2 is 0.2327 for all sixteen features and 0.2354 for a bias term plus the score differential alone, while a tree reaches 0.388. Either rebuild it on rows collected at declaration points as well as ask points and refit it jointly with the policy vector, or retire it and score declarations directly on allocation probability. The measurement decides; the present state — sixteen coefficients carrying one feature's worth of signal — should not survive either way.

The narrow claim, and the broader one. The narrow claim of this report is that a specific, reproducible failure of a specific agent was diagnosed to an arithmetic cause in its ask score, that the cause was removed, that the removal cost nothing measurable in aggregate win rate and left FishBot v0.4 marginally ahead on minimax regret, and that it helped substantially against one class of deceptive opponent while hurting measurably against another. The broader observation is about evaluation. Every statistic in §4 was available to the v0.4 study; none of it was collected, because the panel it was measured against contained no opponent of its own strength, and the failure is a strong-opponent phenomenon. The mirror belongs in the fitting panel, the pathology counters belong in the commit gate, and the per-opponent breakdown with an explicit worst case belongs in the abstract — not because they are more rigorous, but because an aggregate win rate against weaker opponents would have reported all of this as a success.

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